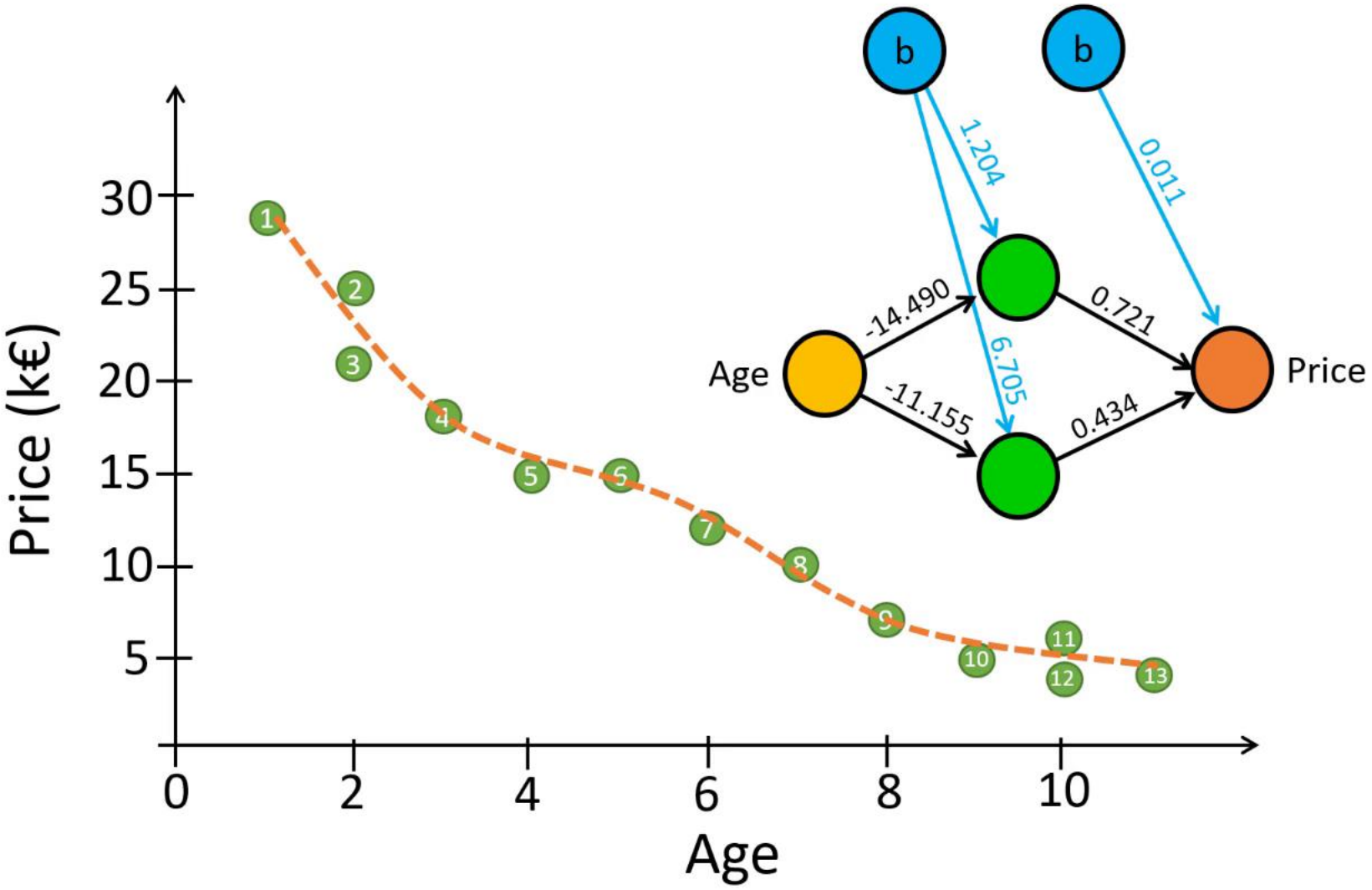
Regression - ANN

Dr.Hafeez Anwar

ANN vs Regression

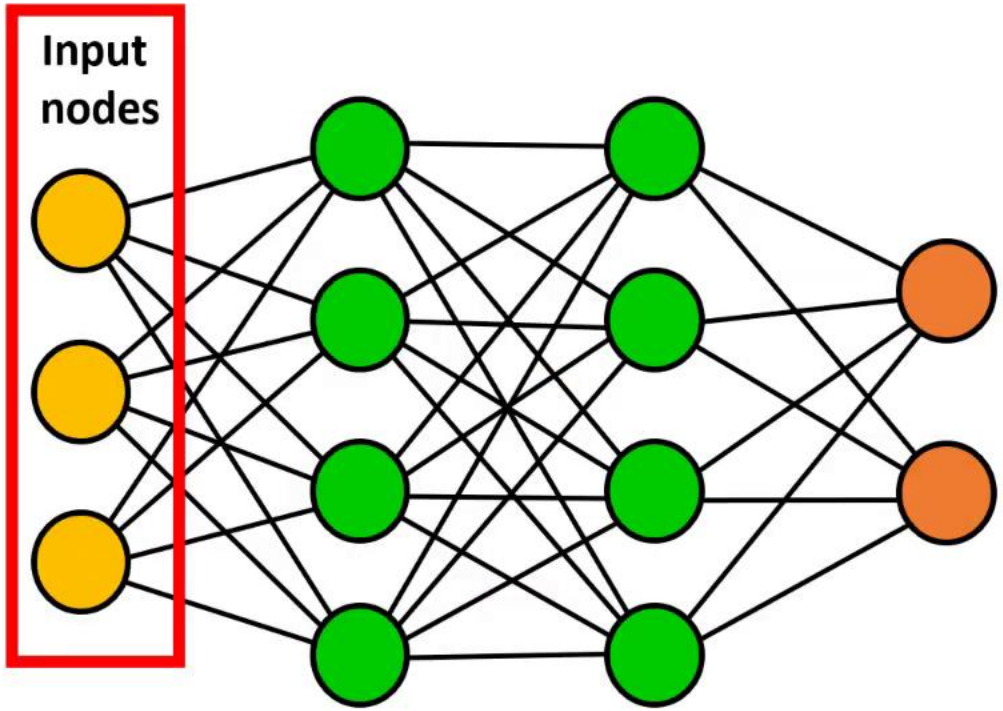


Car	Price (k€)	Age (years)
1	29	1
2		
3		
4	18	3
5	15	4
6	15	5



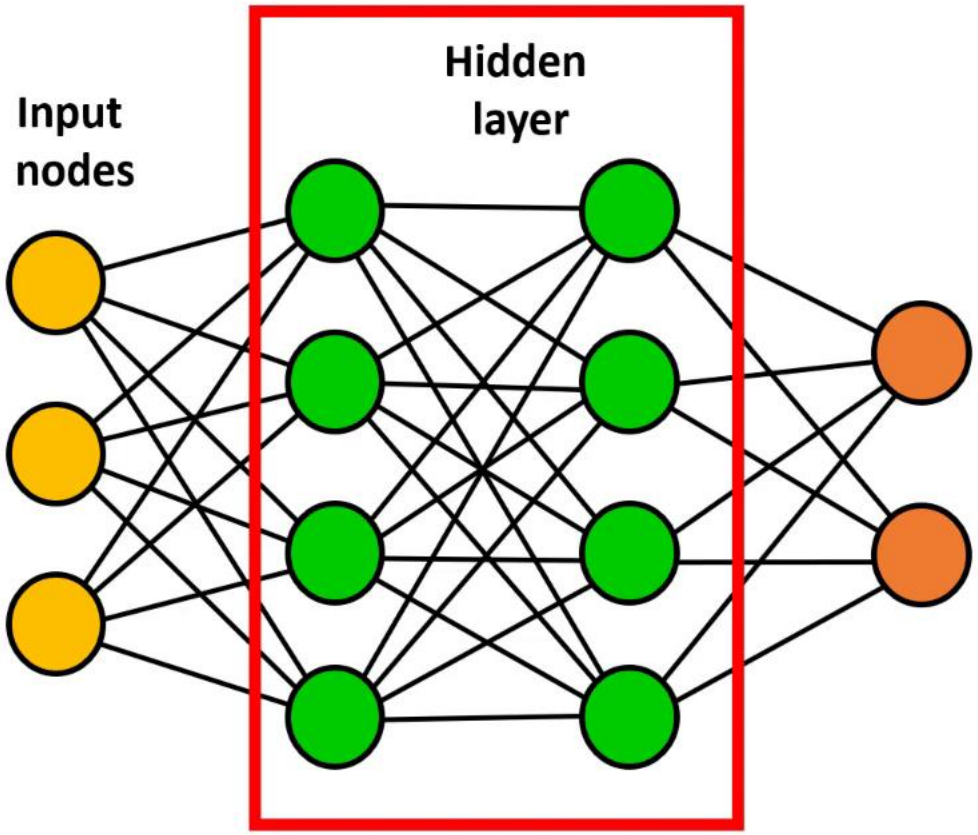
we will discuss the similarities and differences between neural networks and regression.

ANN



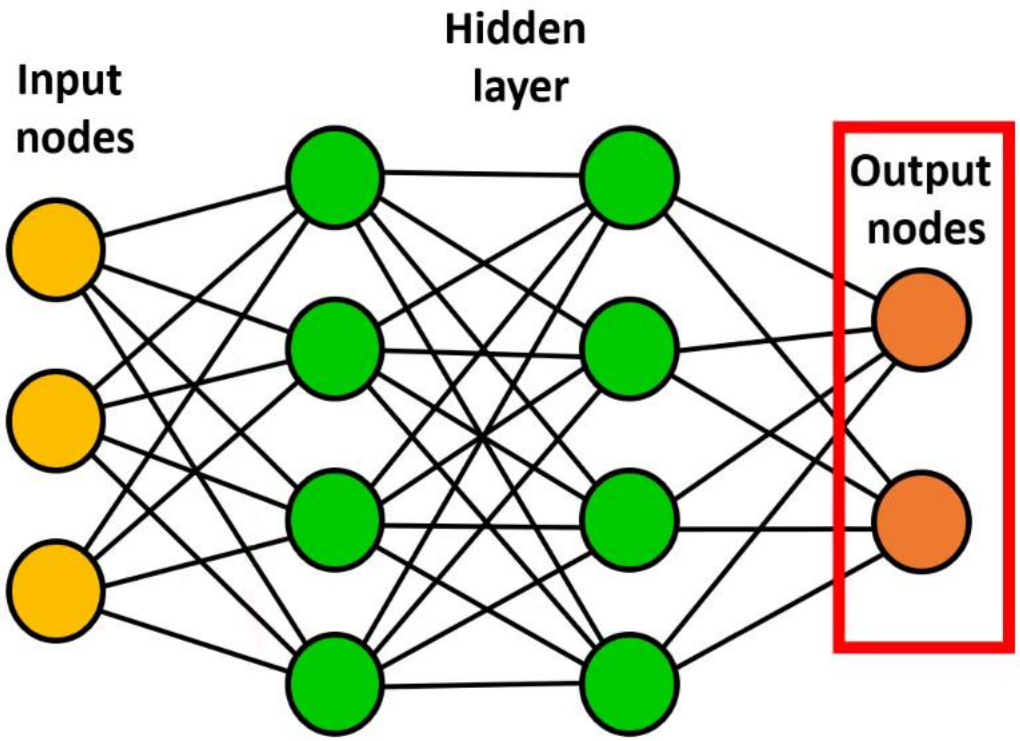
A neural network consists of input nodes,

ANN



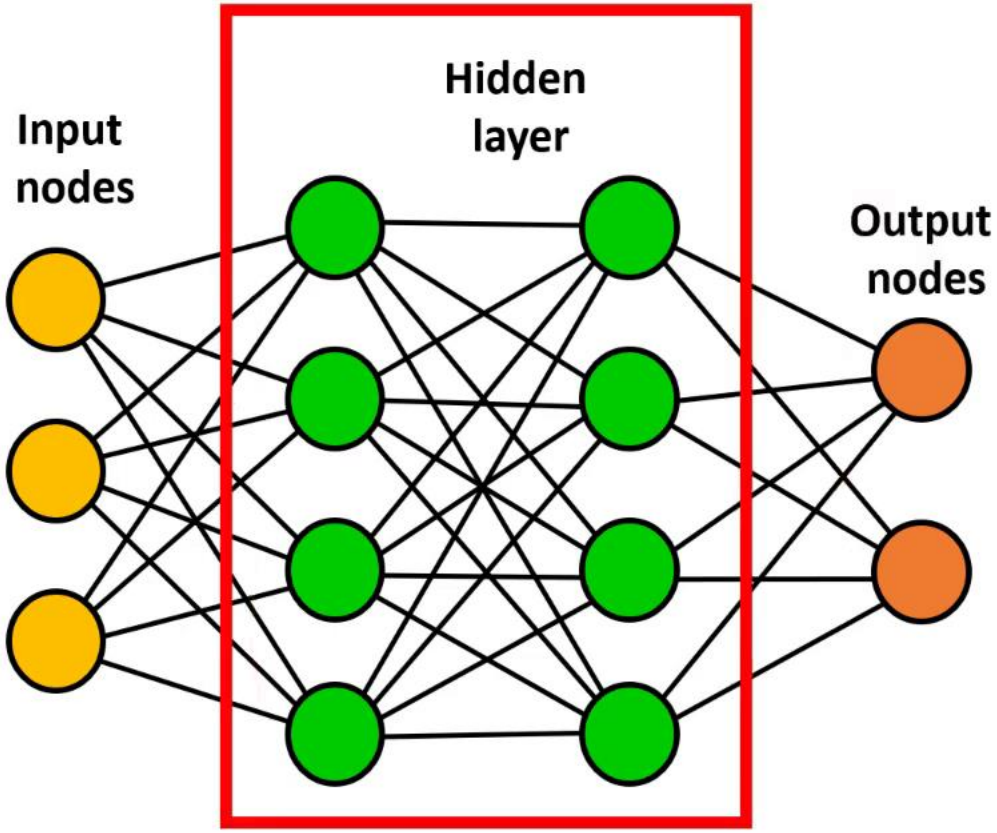
a hidden layer,

ANN



and output nodes.

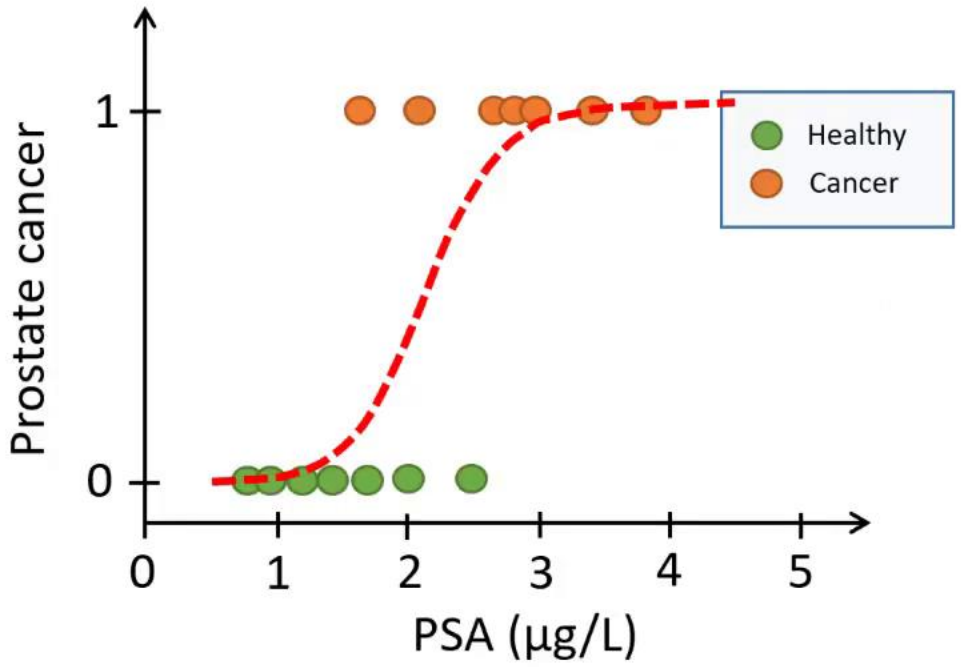
ANN

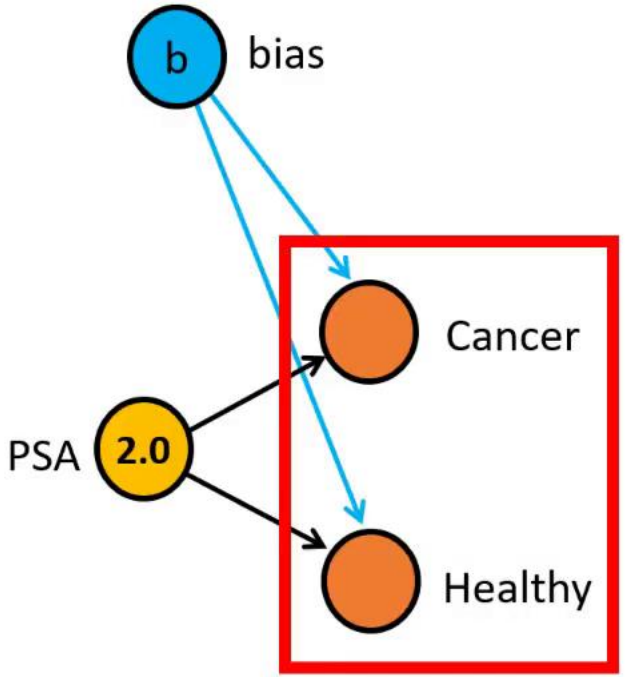


To compare neural networks with regression, we will first look at networks without a hidden layer, and then see what happens when we include the hidden layer.

ANN vs Logistic regression

$$f(z) = \frac{1}{1 + e^{-z}}$$





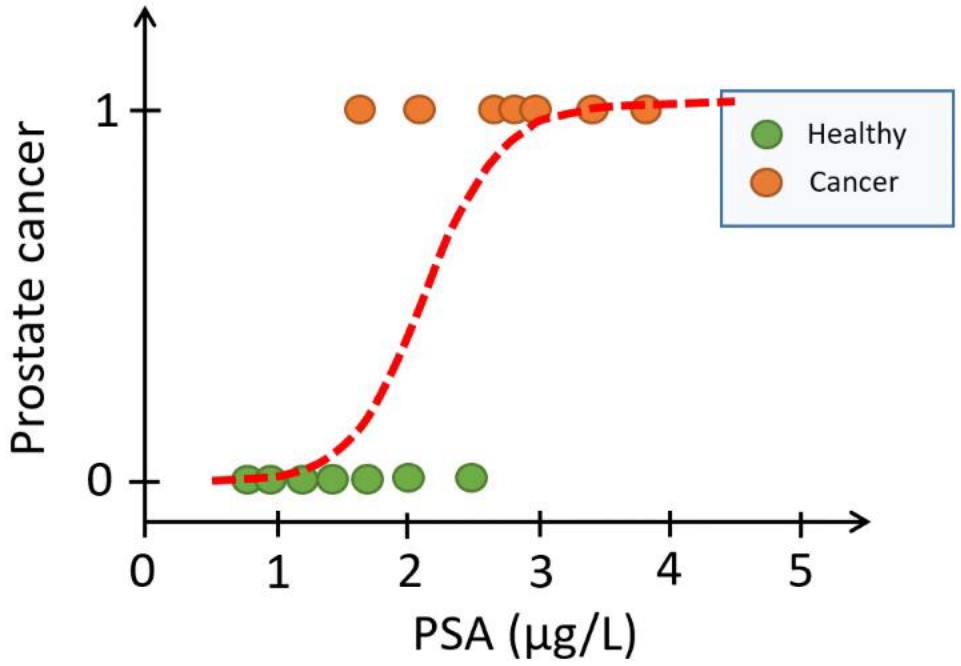
Logistic regression

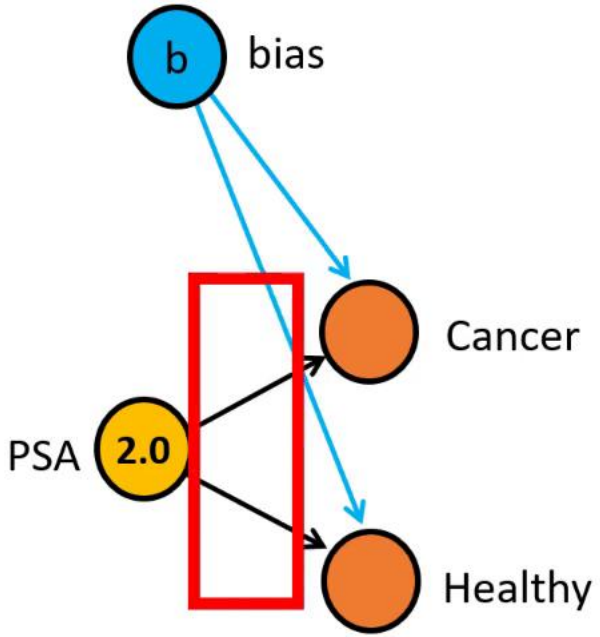
$$P(cancer) = \frac{1}{1 + e^{-(b_0 + b_1 PSA)}}$$

we saw that a network with a binary outcome,

ANN vs Logistic regression

$$f(z) = \frac{1}{1 + e^{-z}}$$





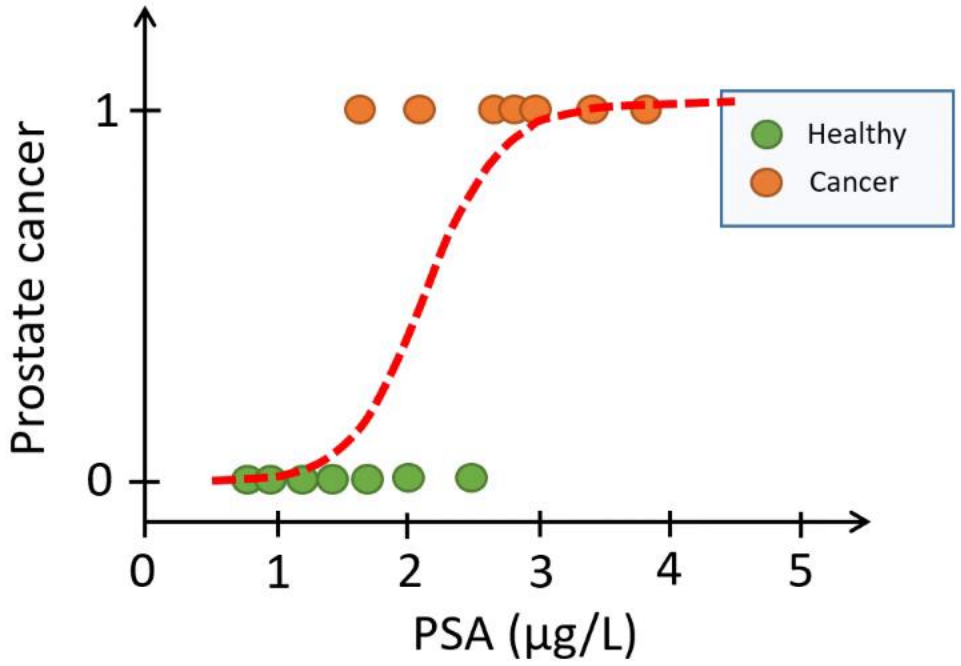
Logistic regression

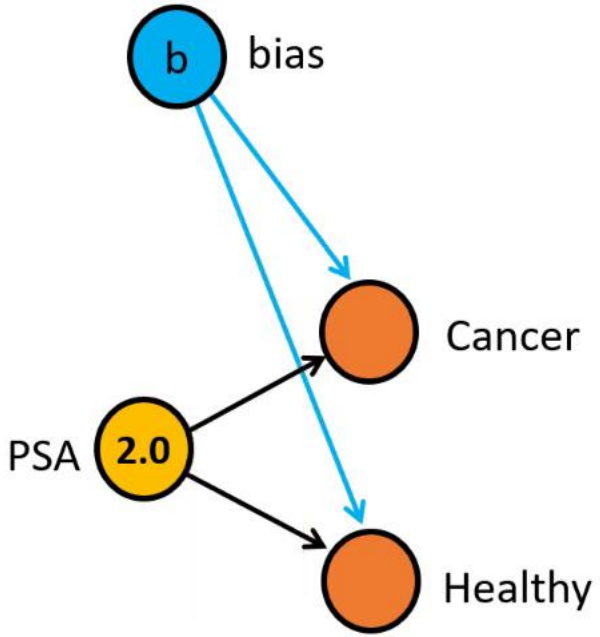
$$P(cancer) = \frac{1}{1 + e^{-(b_0 + b_1 PSA)}}$$

with no hidden layer,

ANN vs Logistic regression

$$f(z) = \frac{1}{1 + e^{-z}}$$





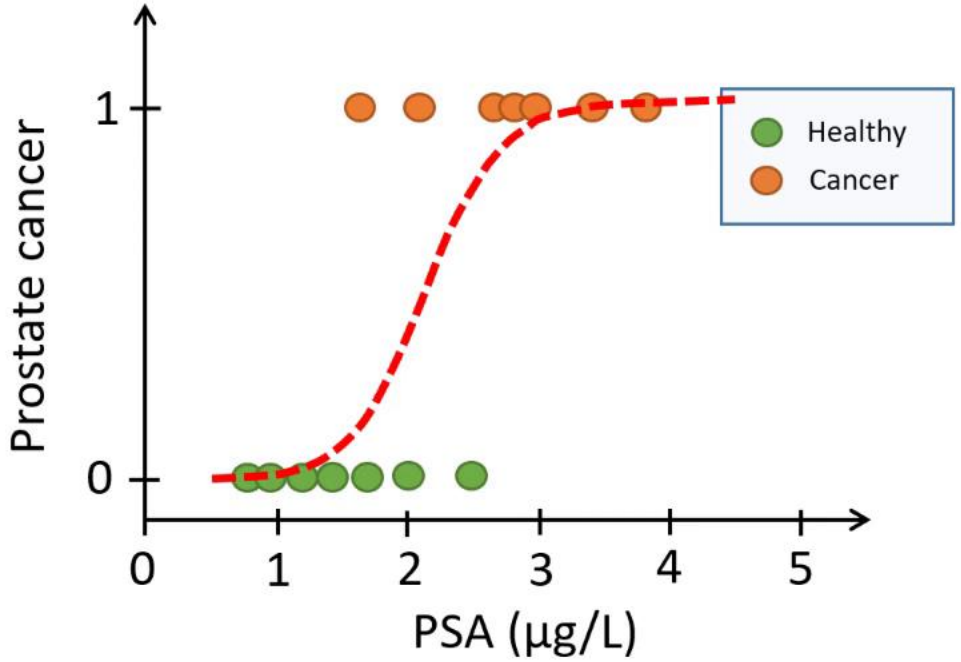
Logistic regression

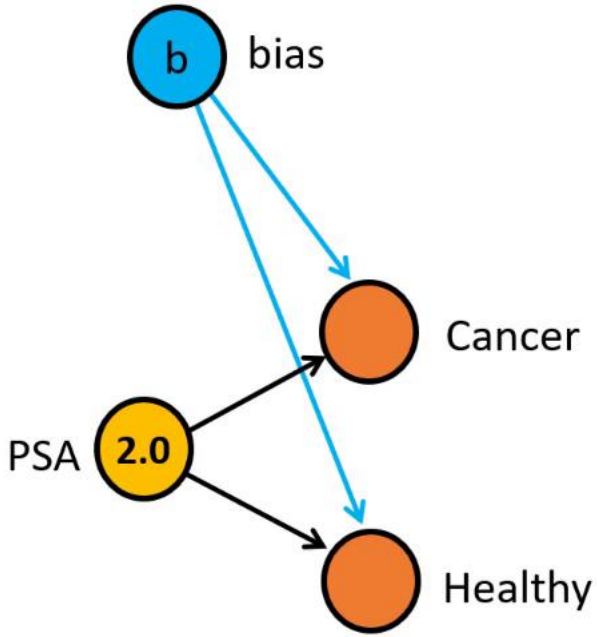
$$P(cancer) = \frac{1}{1 + e^{-(b_0 + b_1 PSA)}}$$

and with a logistic activation function,

ANN vs Logistic regression

$$f(z) = \frac{1}{1 + e^{-z}}$$



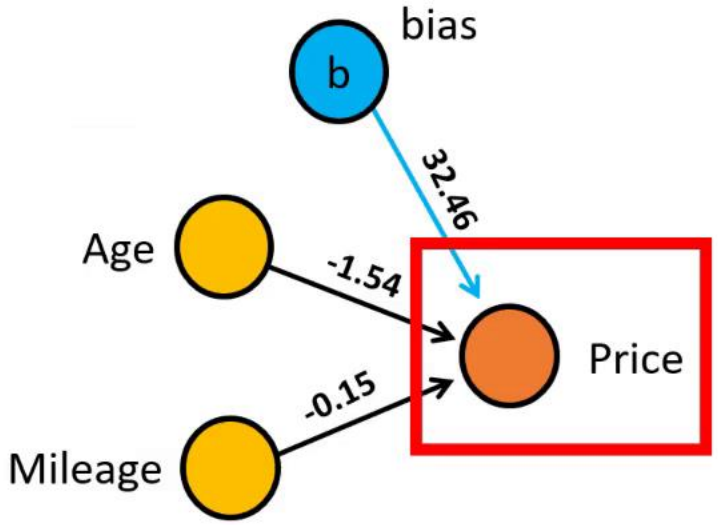


Logistic regression

$$P(cancer) = \frac{1}{1 + e^{-(b_0 + b_1 PSA)}}$$

generated the exact same results as logistic regression.

ANN vs Linear regression





we will instead focus on neural networks with a single output node with a continuous outcome, such as the predicted price of a car.

ANN vs Linear regression

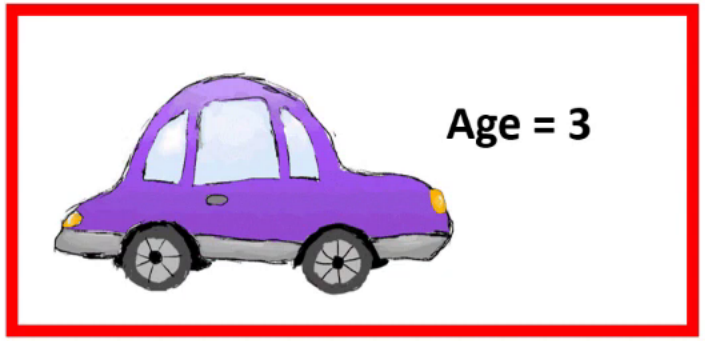
Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



Let’s start to compare a neural network with simple linear regression on the following data.

ANN vs Linear regression

Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



Suppose that we like to predict the price of the following car that is three years old.

ANN vs Linear regression

Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



Age = 3

To predict the price of this car, we collect sales data on six similar cars.

ANN vs Linear regression

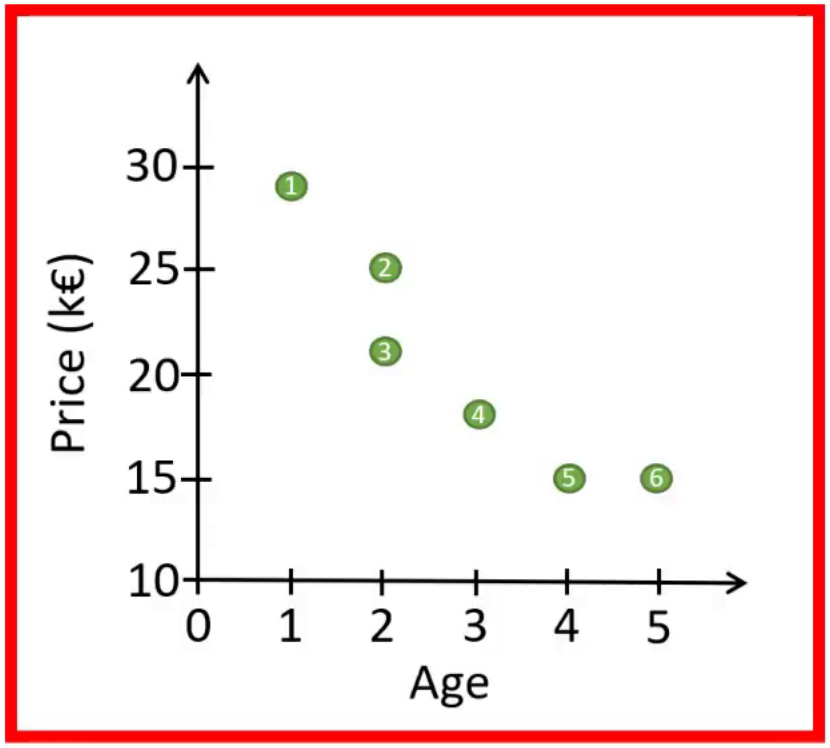
Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



Age = 3

For example, car number three is two years old and has a price of 21.000 euros.

ANN vs Linear regression



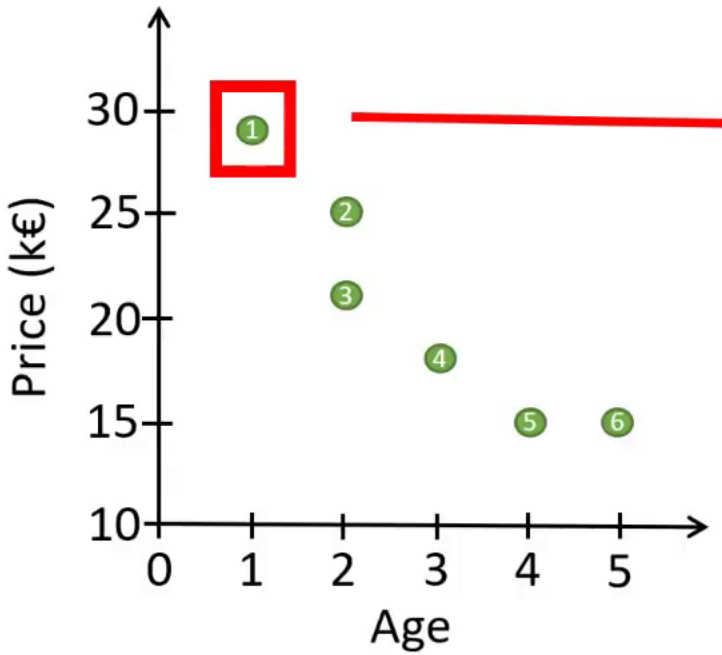
Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



Age = 3

Let's plot this data,

ANN vs Linear regression



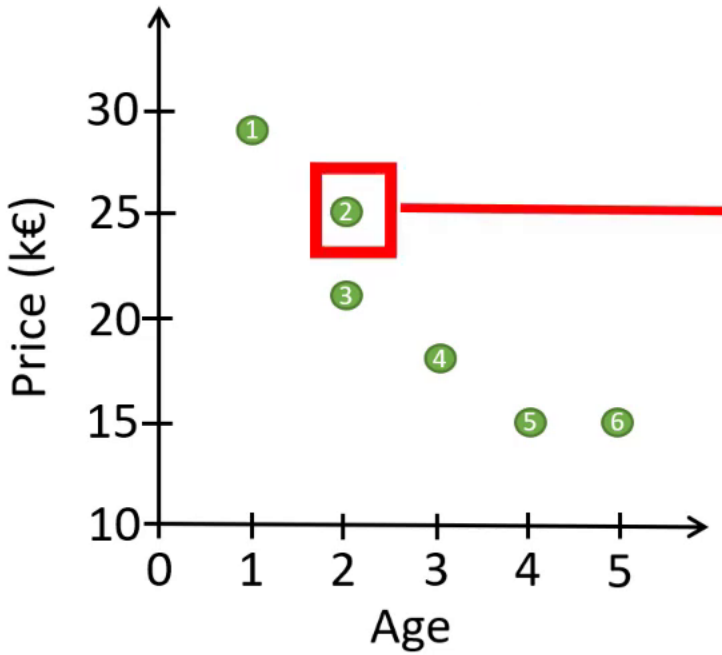
Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



Age = 3

where this data point represents car number one,

ANN vs Linear regression



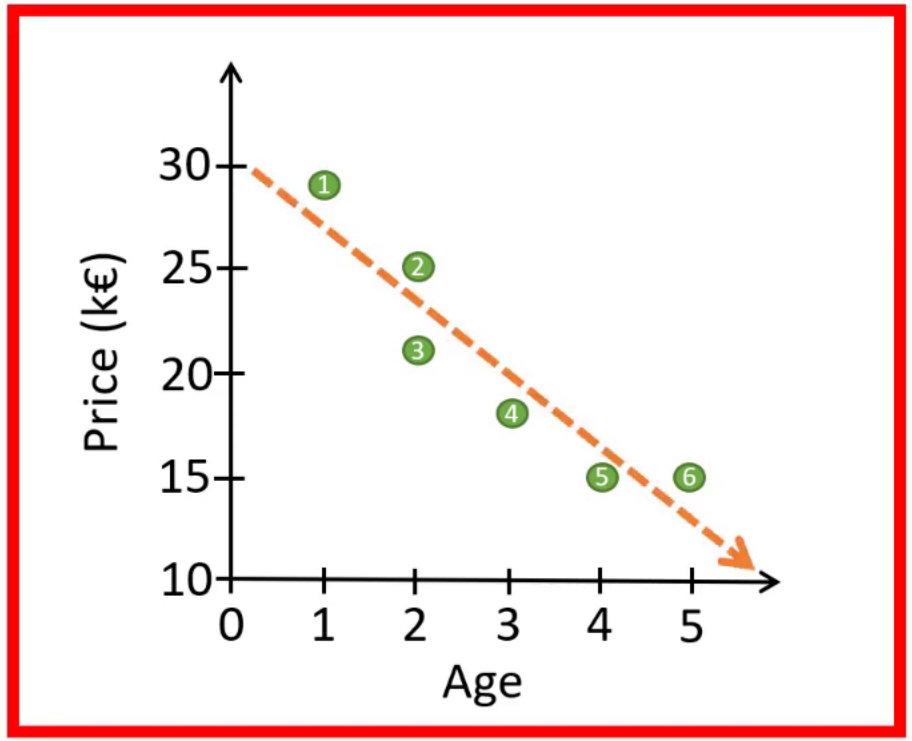
Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



Age = 3

whereas this point represents car number two, and so forth.

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5

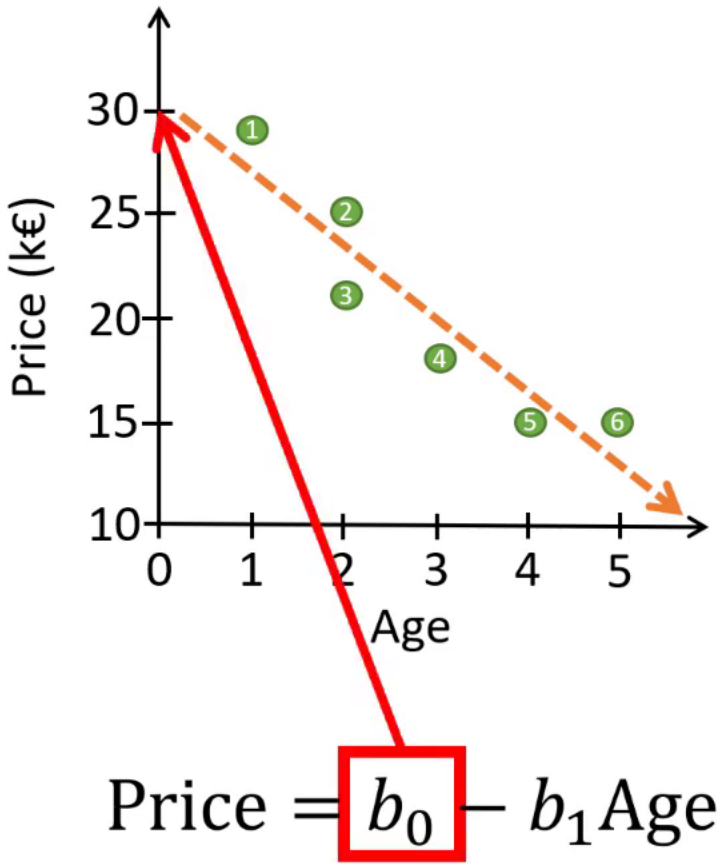
$$\text{Price} = b_0 - b_1 \text{Age}$$



Age = 3

Then we can use simple linear regression to fit a straight line to this data so that the line is as close as possible to the data points.

ANN vs Linear regression



$$\text{Price} = b_0 - b_1 \text{Age}$$

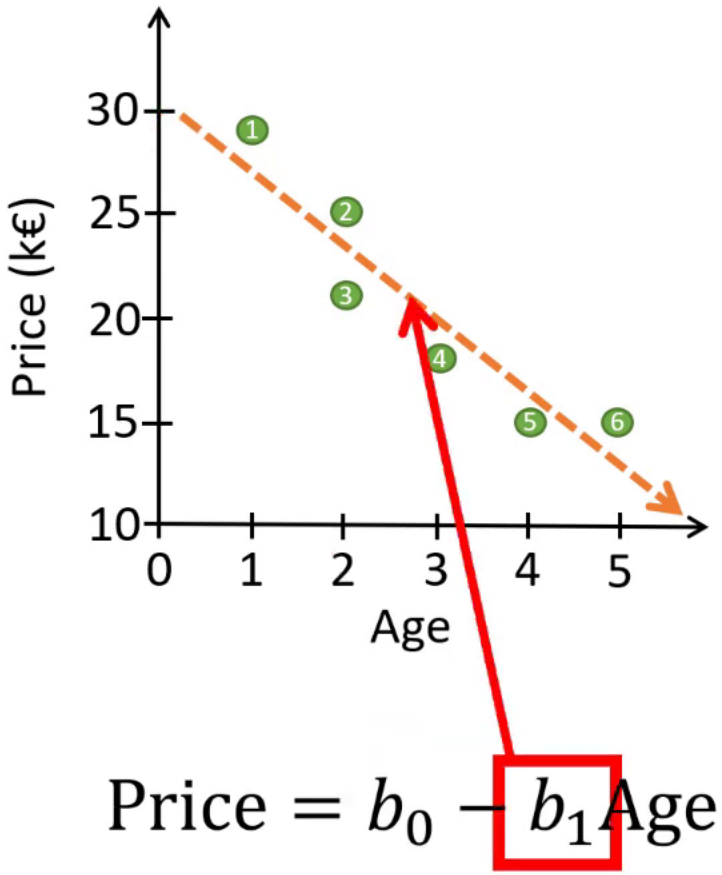
Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



Age = 3

This means that we try to find the optimal value of the intercept,

ANN vs Linear regression



$$\text{Price} = b_0 - b_1 \text{Age}$$

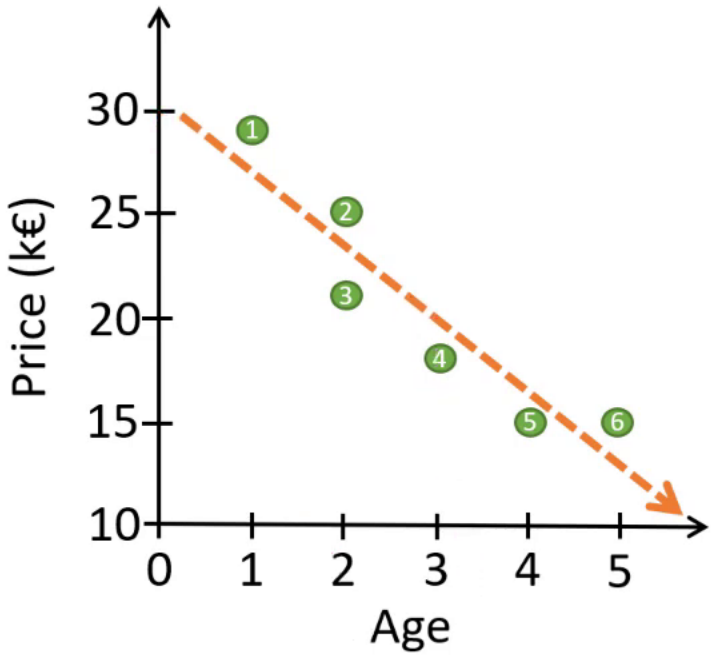
Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



Age = 3

and the slope,

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

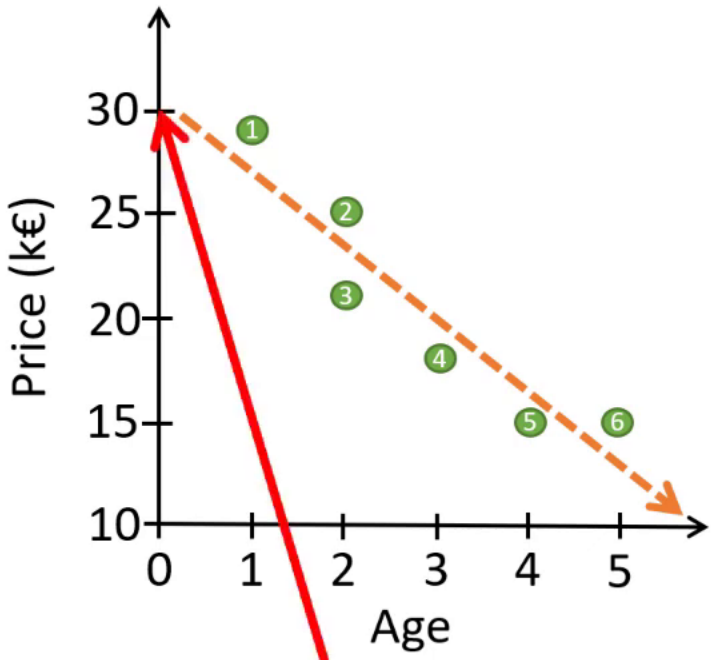
$$\text{Price} = b_0 - b_1 \text{Age}$$



Age = 3

so that the sum of the squared residuals, or errors, is as small as possible

ANN vs Linear regression



$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$

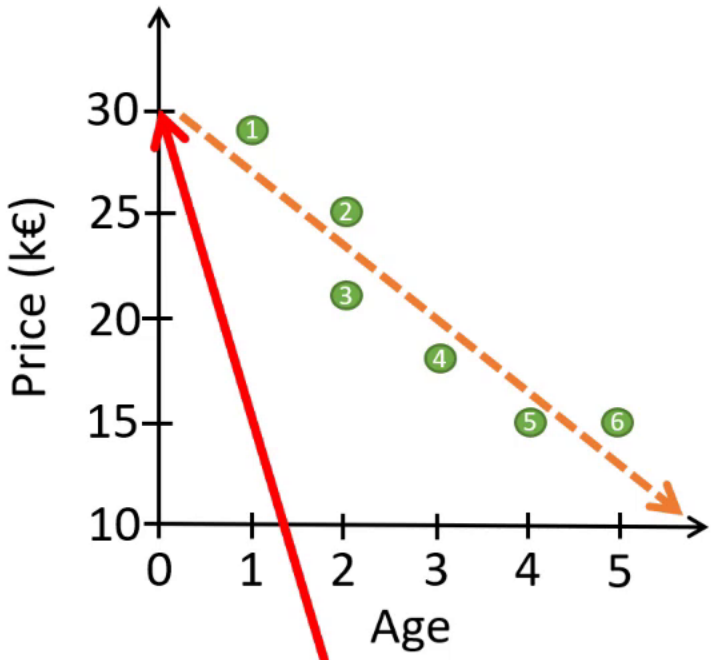
Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



Age = 3

For this data, the optimal value of the intercept is about 30.57,

ANN vs Linear regression



$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$

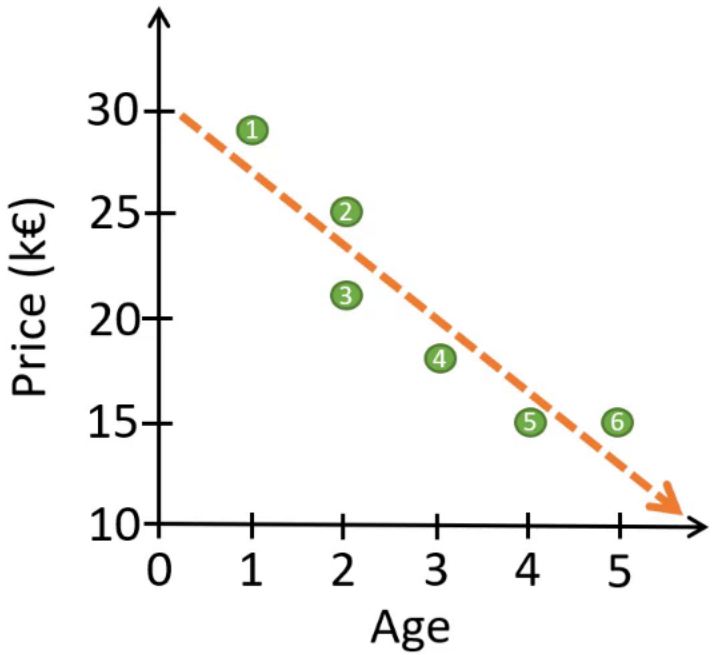
Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



Age = 3

which means that the predicted price of a brand new car is 30.570 euros.

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5

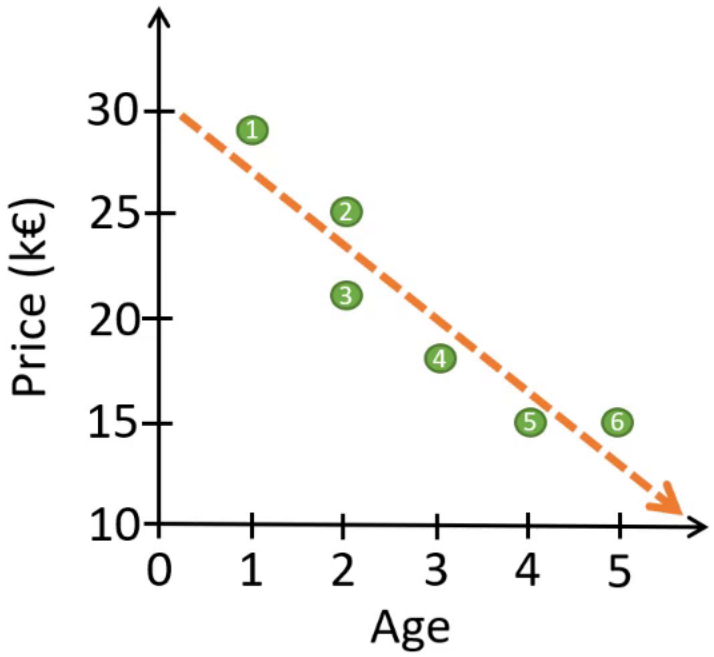
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



Age = 3

The optimal value of the slope of the line has been estimated to negative 3.55, which can be interpreted as the price is predicted to decline by 3550 euros when the car gets one year older.

ANN vs Linear regression



$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$

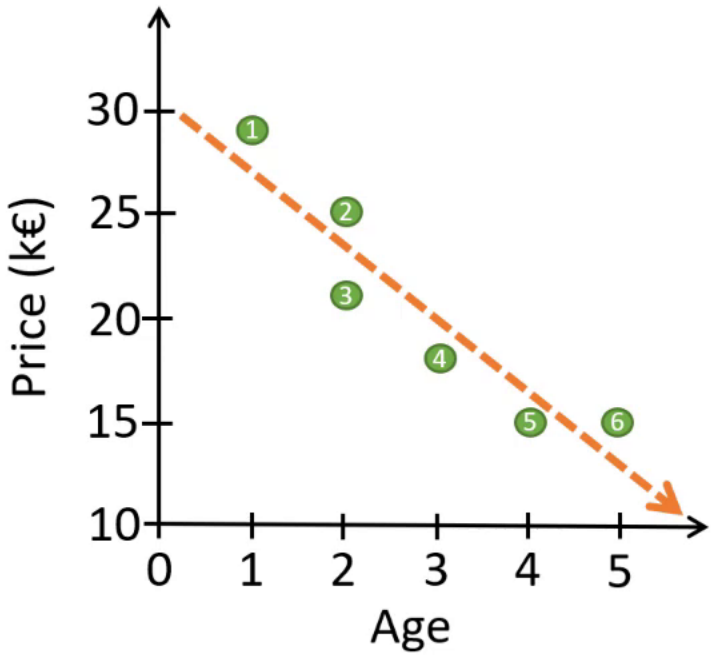
Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



Age = 3

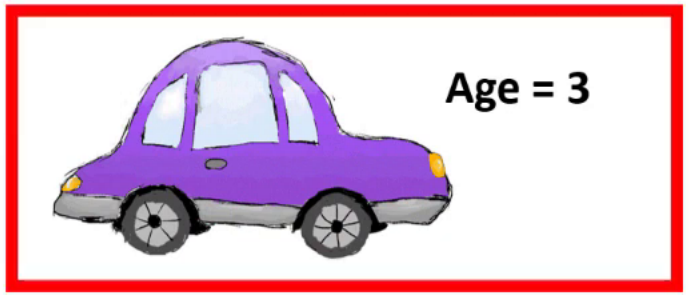
Once we have fitted the line to the data and estimated the parameters of the model,

ANN vs Linear regression



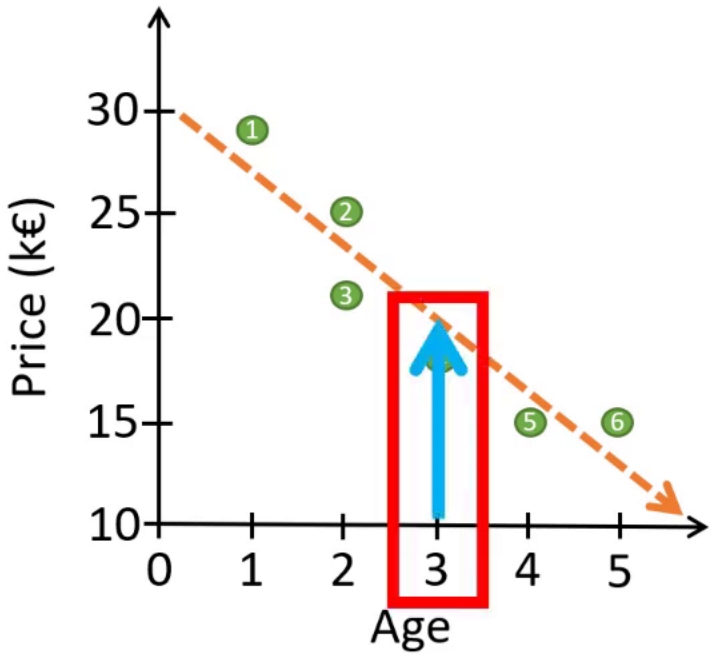
Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5

$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



we can predict the price of our car with an age of 3.

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5

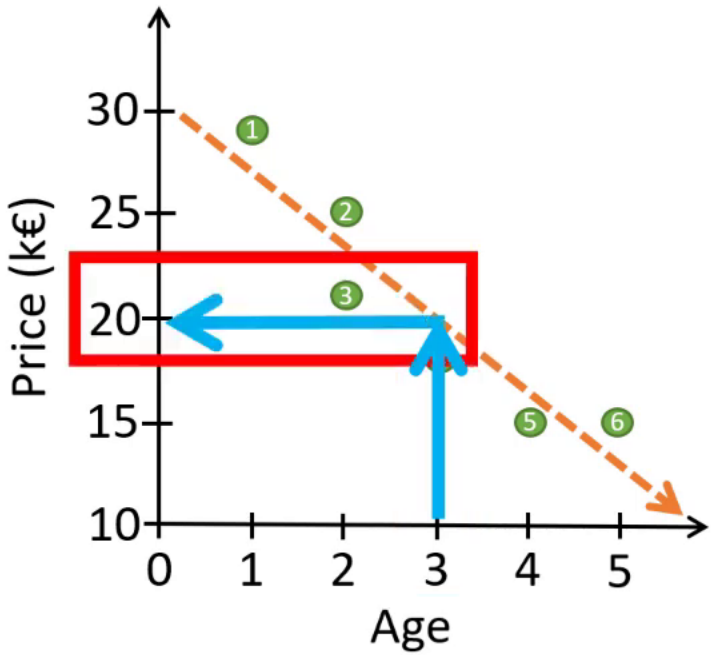
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



Age = 3

We can draw a vertical line from three until we hit the regression line,

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5

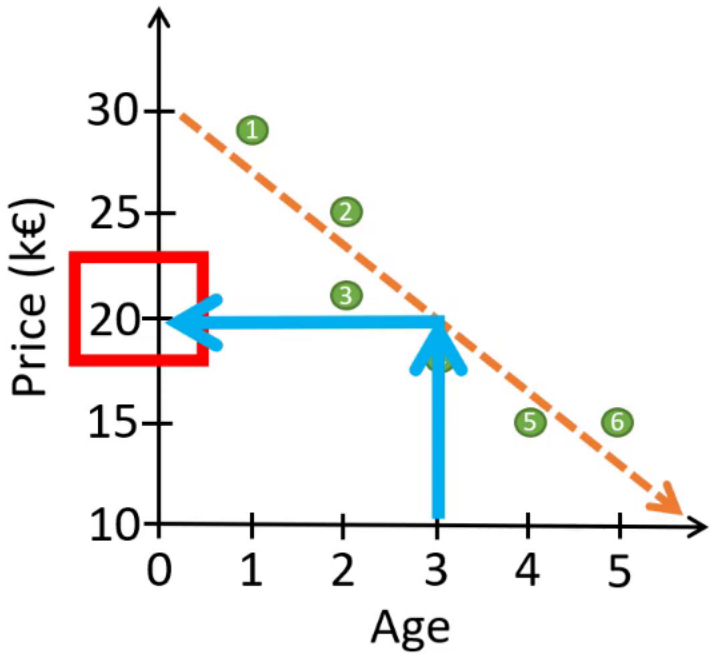
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



Age = 3

and then we draw a horizontal line until we hit the y-axis.

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5

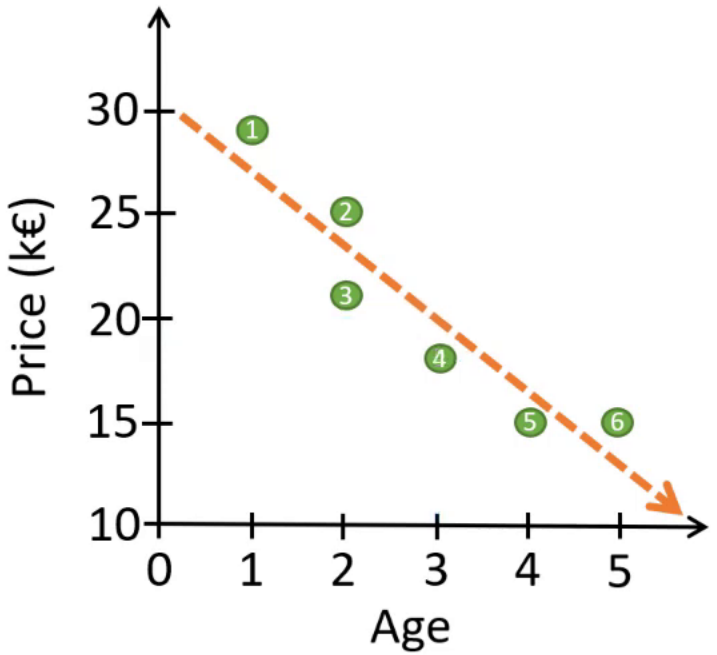
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



Age = 3

The predicted price of the car, according to the model, is about 20.000 euros.

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5

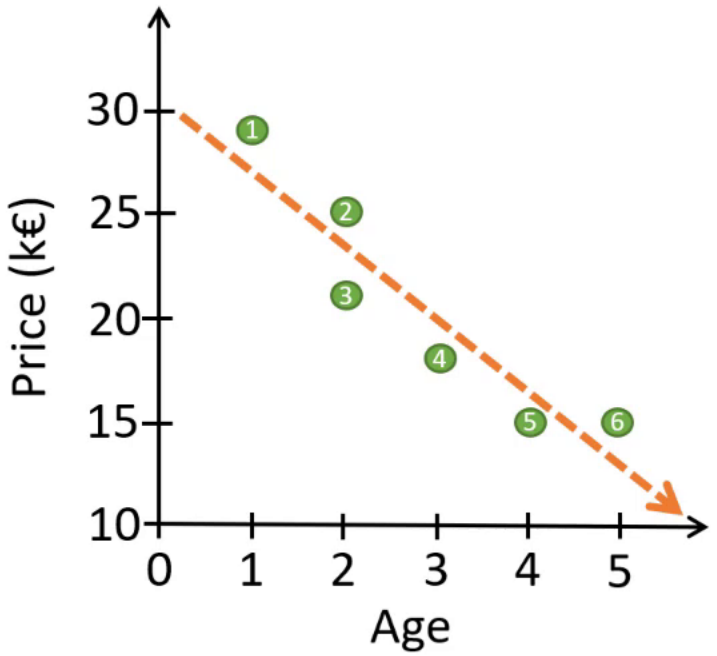
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



Age = 3

We can compute a more accurate prediction if we instead use the equation,

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5

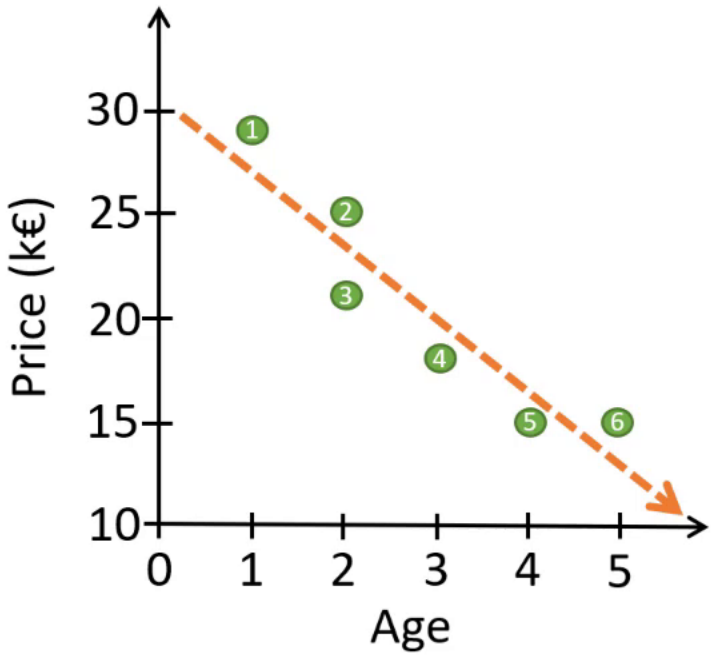
$$\text{Price} = 30.57 - 3.55 \cdot 3$$



Age = 3

where we simply set the age to three,

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5

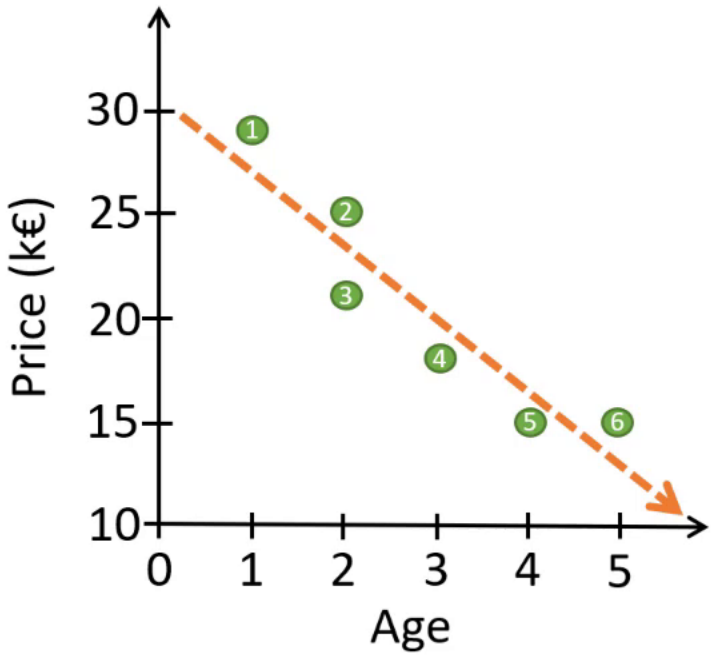
$$\text{Price} = 30.57 - 3.55 \cdot 3 = 19.92$$



Age = 3

and do the math.

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5

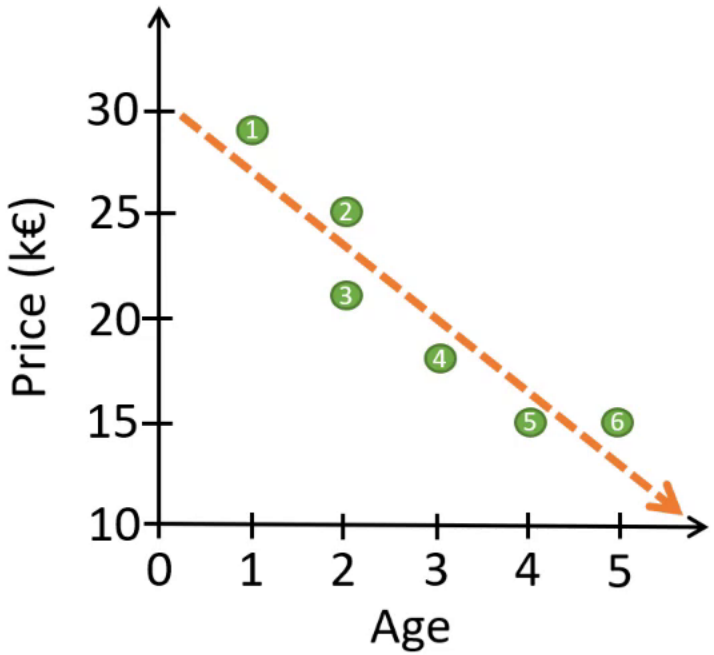
$$\text{Price} = 30.57 - 3.55 \cdot 3 = 19.92$$



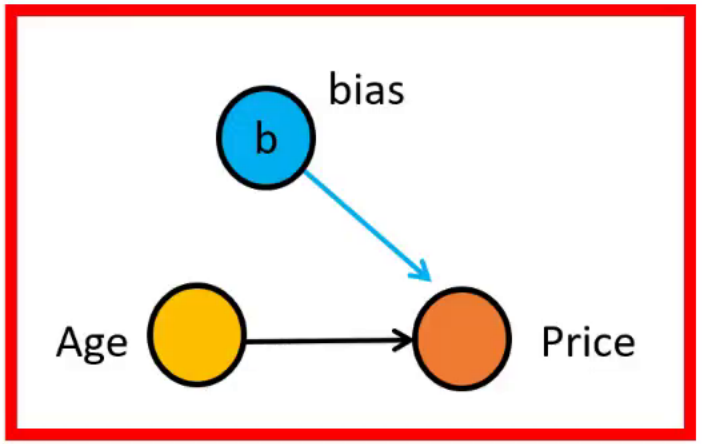
Age = 3

By using linear regression, we have estimated the price of this car to about 19.920 euros.

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



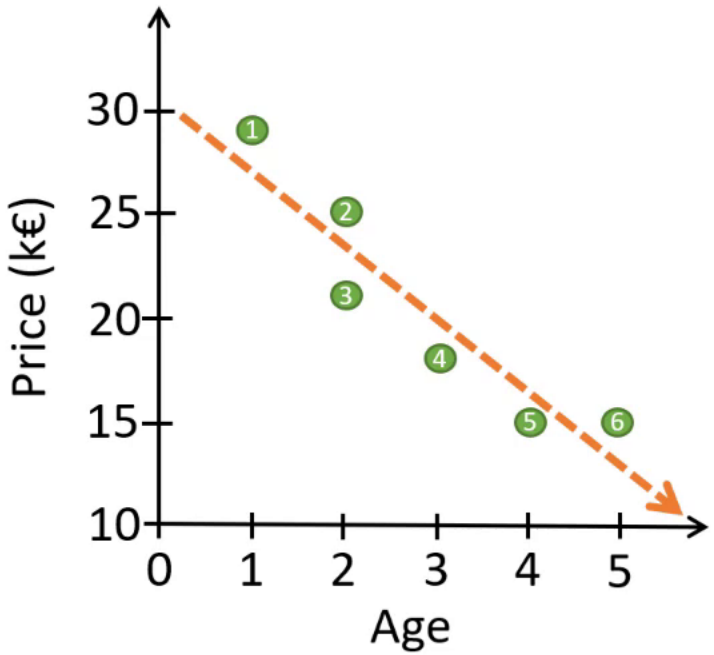
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



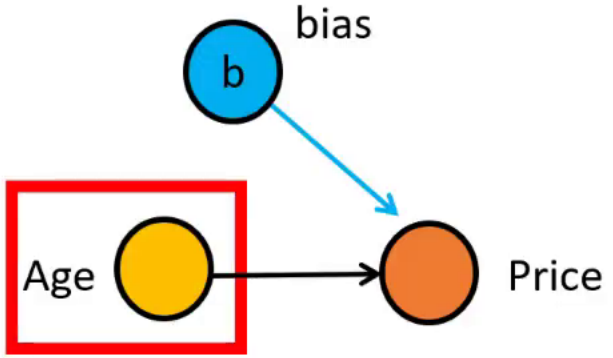
Age = 3

If we would predict the same thing with a network,

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



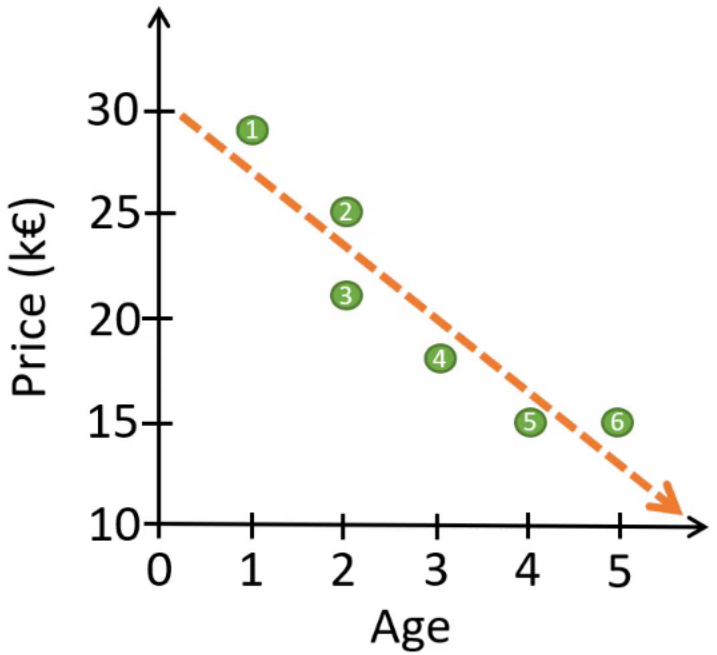
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



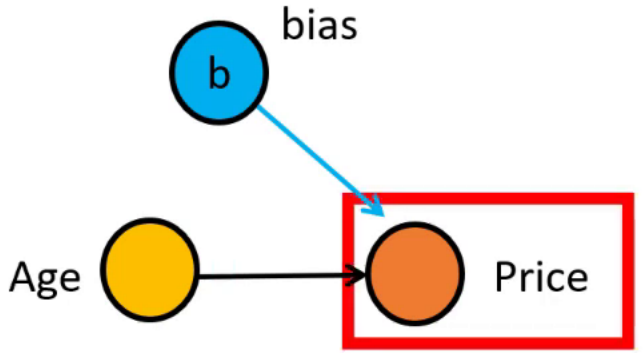
Age = 3

we would use only one input node,

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



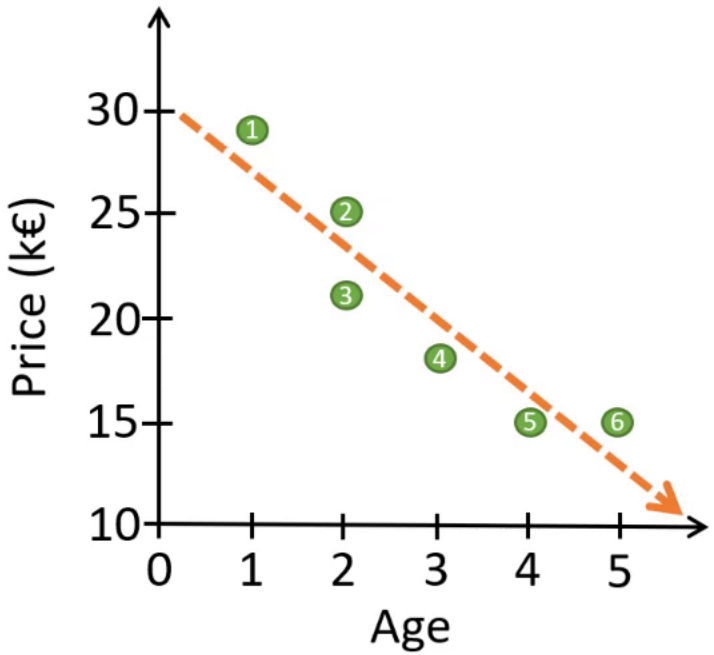
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



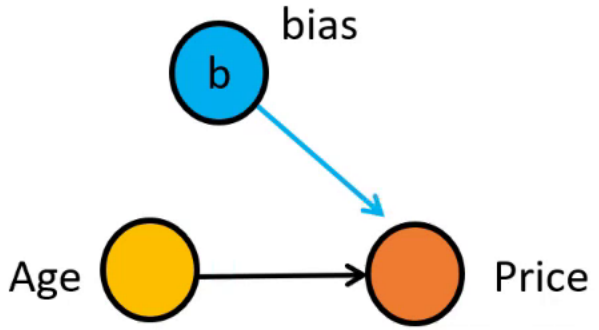
Age = 3

and one output node,

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



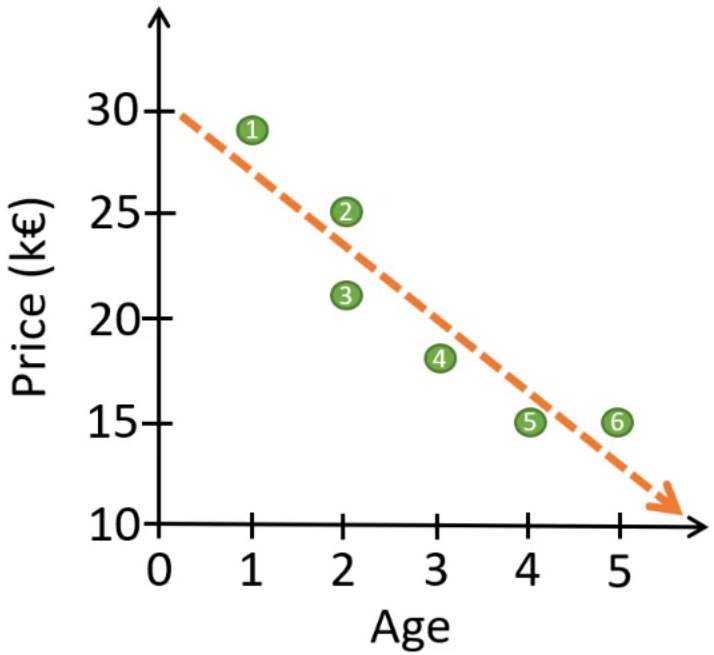
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



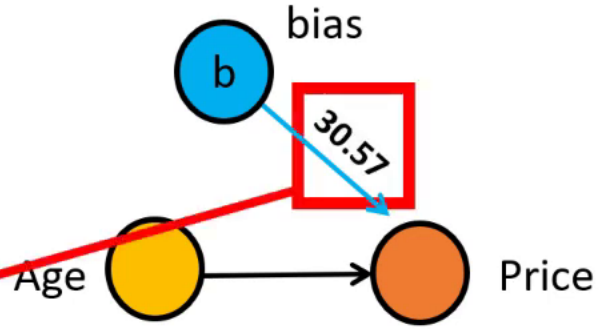
Age = 3

and train this network based on the following data.

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



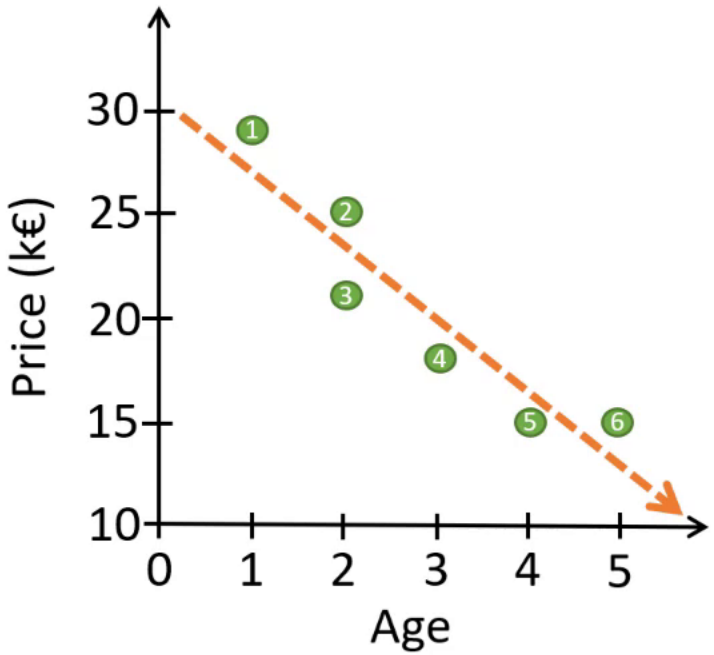
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



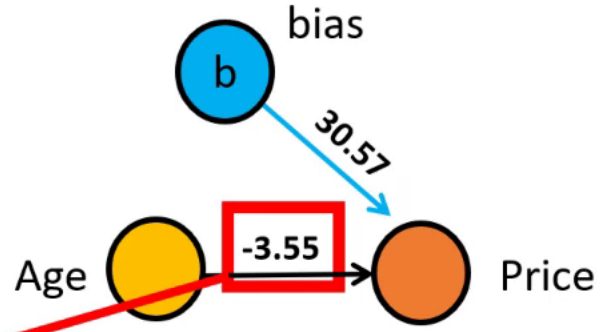
Age = 3

The weight associated with the bias would be estimated to the same value as the intercept in the regression model,

ANN vs Linear regression



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5



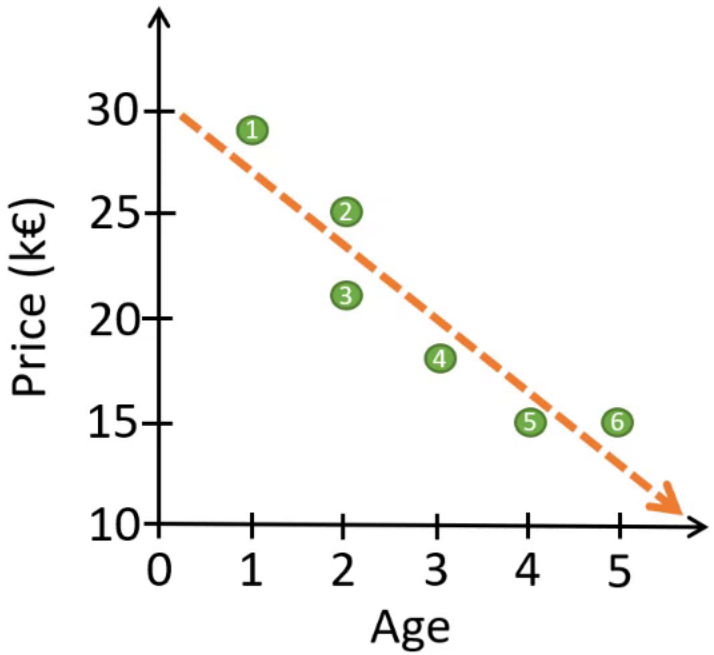
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



Age = 3

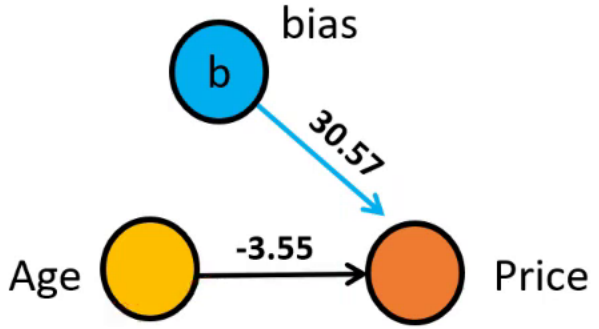
whereas this weight will correspond to the slope of the regression model.

ANN vs Linear regression



$$z = w_0 + \sum_{i=1}^{n} w_i \, x_i$$

$$f(z) = z$$



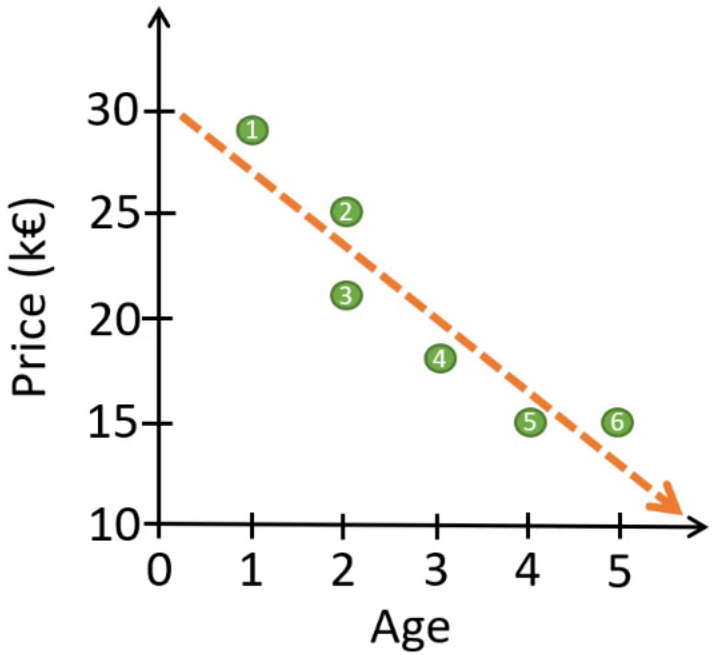
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



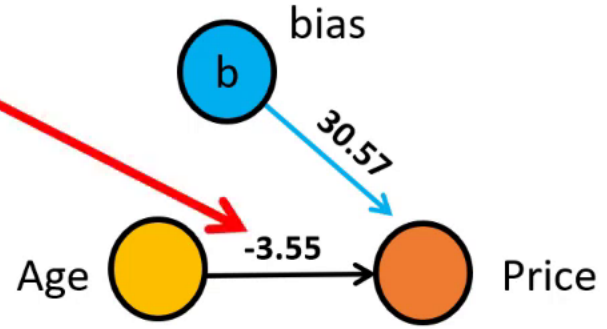
Age = 3

Remember that when we train a network, we try to find the optimal weights, that for example result in the lowest possible sum of squared errors, which is the same thing as we do in linear regression.

ANN vs Linear regression



$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$



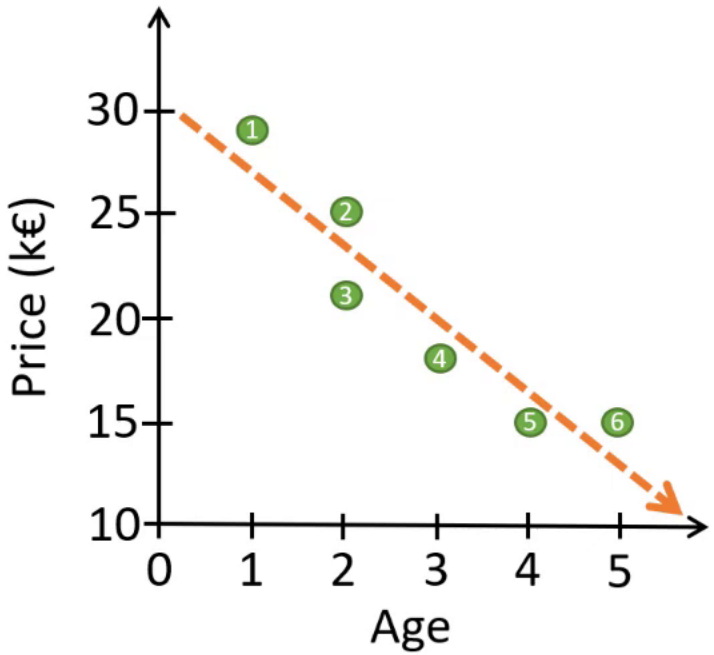
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



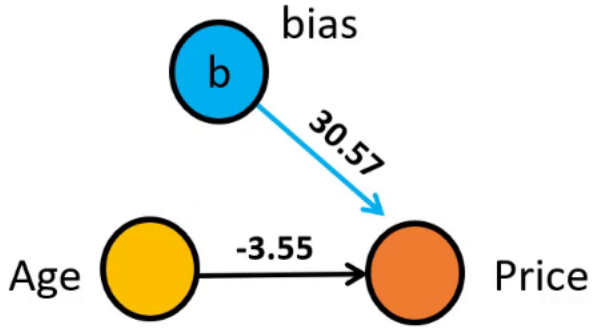
Age = 3

Since we in this example only have one input signal and no hidden layer, n is here equal to one,

ANN vs Linear regression



$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$



$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



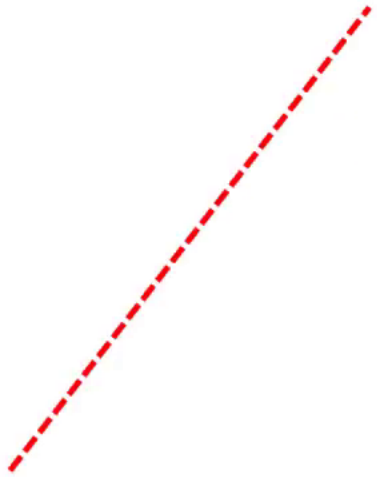
Age = 3

which means that we can delete this from the equation,

Activation functions

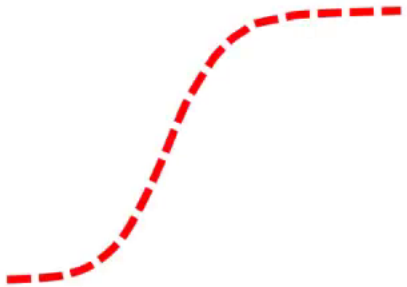
Identity

$$f(z) = z$$



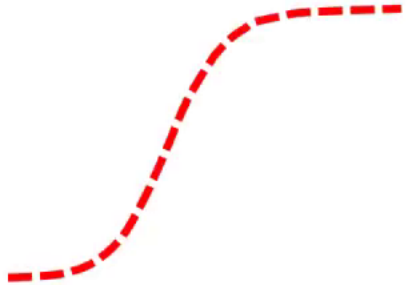
Sigmoid

$$f(z) = \frac{1}{1 + e^{-z}}$$



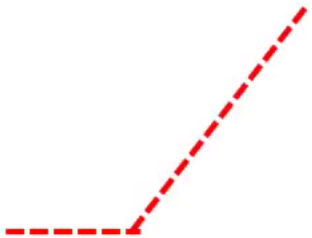
Tanh

$$f(z) = \tanh(z)$$



ReLU

$$f(z) = max(0, z)$$

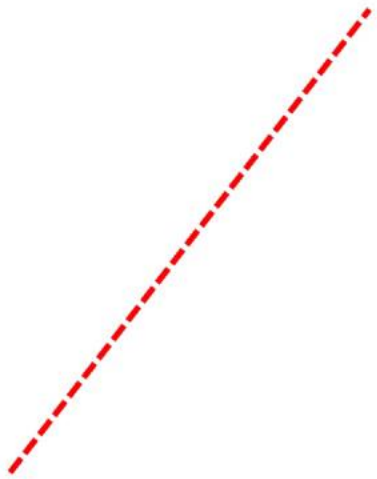


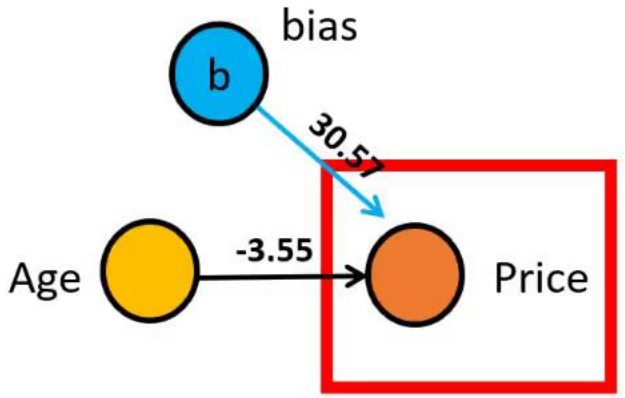
Remember that there are several activation functions to select between for our neural network.

Activation functions

Identity

$$f(z) = z$$



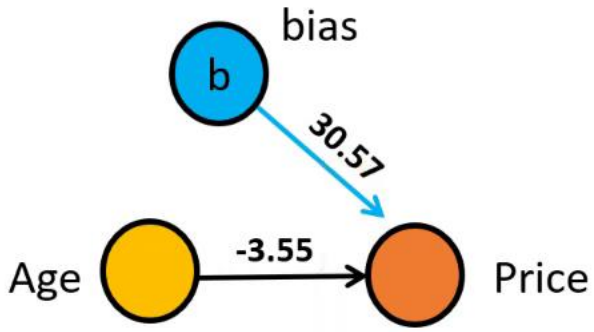


However, when we have a single output node on continuous scale,

Activation functions

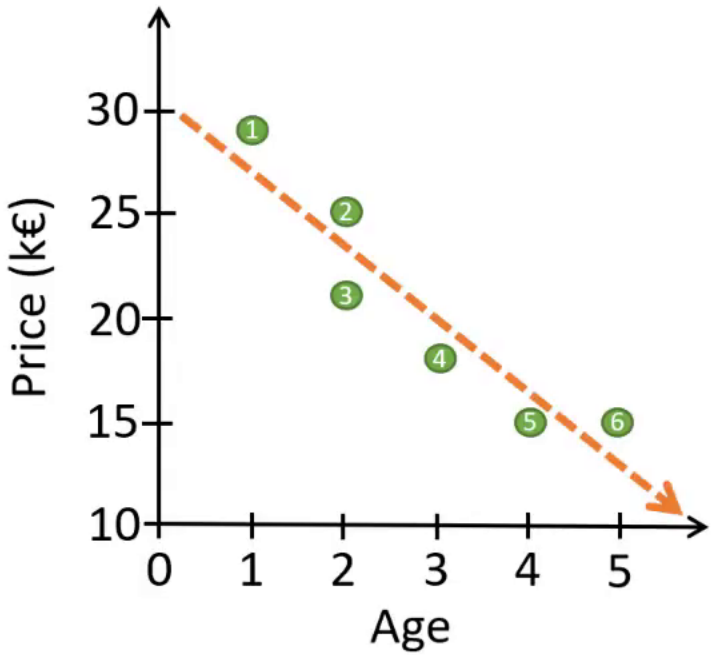
Identity

$$f(z) = z$$



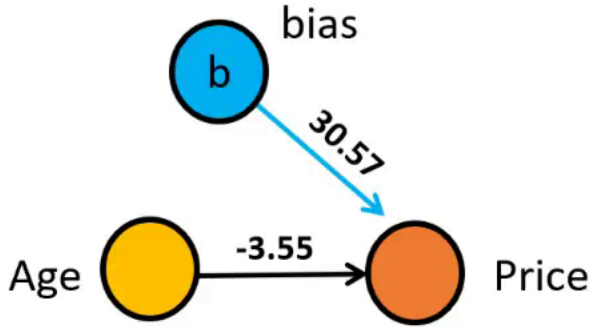
we usually use the identity activation function.

ANN vs Linear regression



$$z = w_0 + w_1 \cdot x_1$$

$$f(z) = z$$



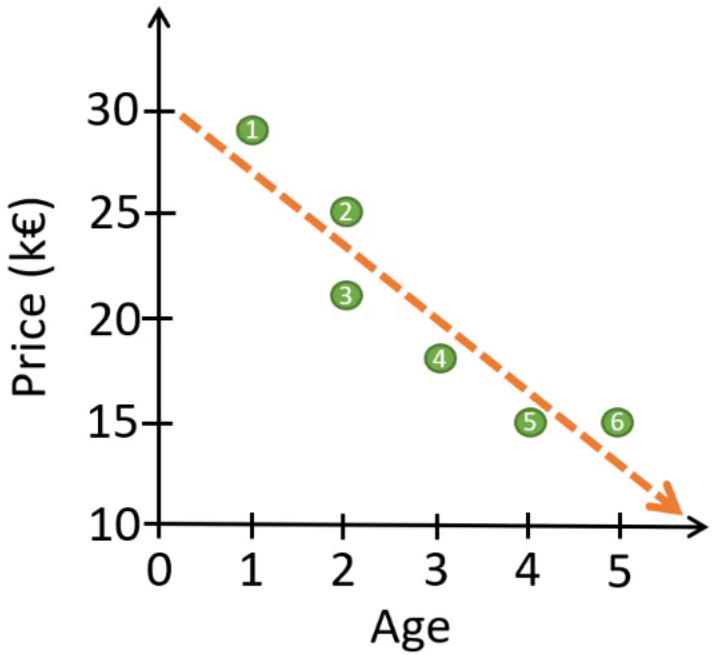
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



Age = 3

If we plug in the equation with the weights in the identity activation function,

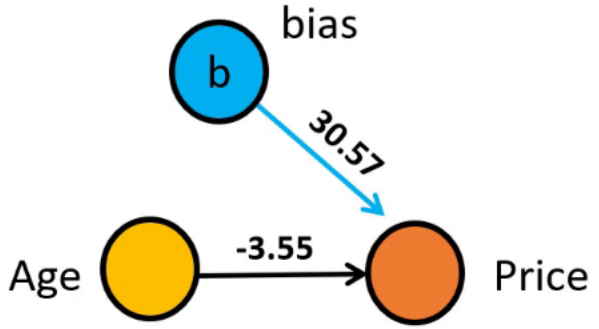
ANN vs Linear regression



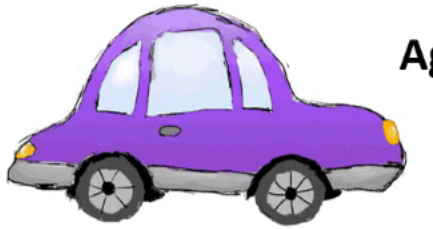
$$z = w_0 + w_1 \cdot x_1$$

$$f(z) = z$$

$$f(z) = w_0 + w_1 \cdot x_1$$



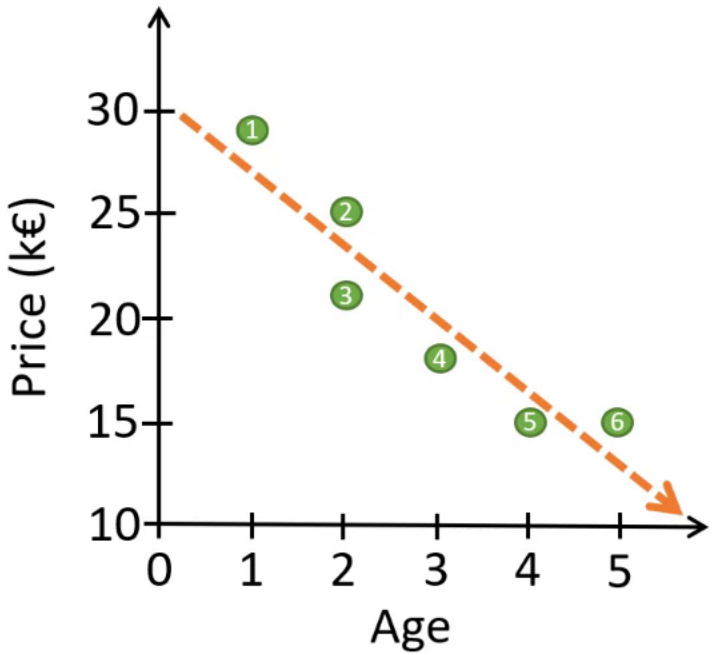
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



Age = 3

we will have the following function that is used to compute the value of the output node.

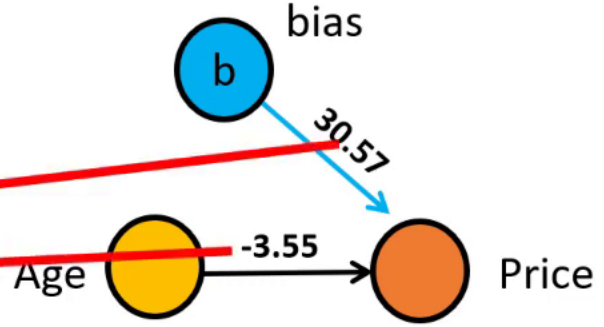
ANN vs Linear regression



$$z = w_0 + w_1 \cdot x_1$$

$$f(z) = z$$

$$f(z) = w_0 + w_1 \cdot x_1$$



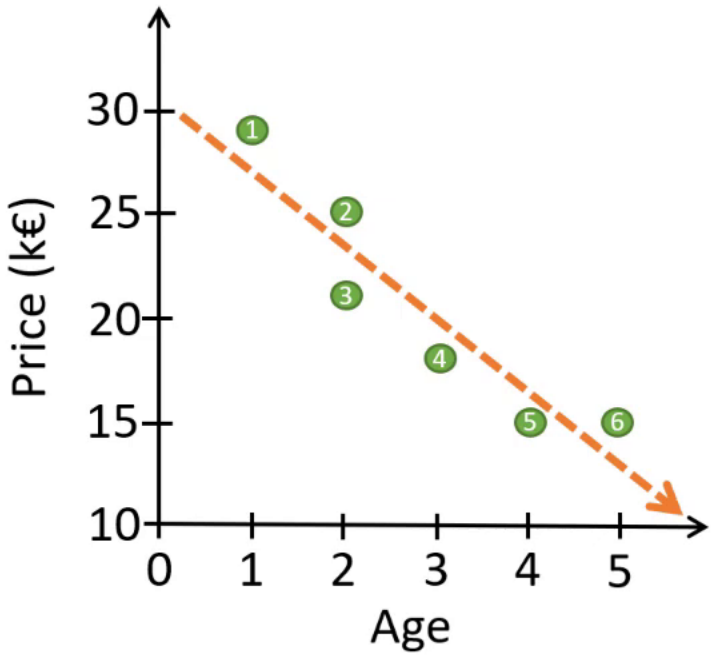
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



Age = 3

Let's also plug in the values of the weights in the function,

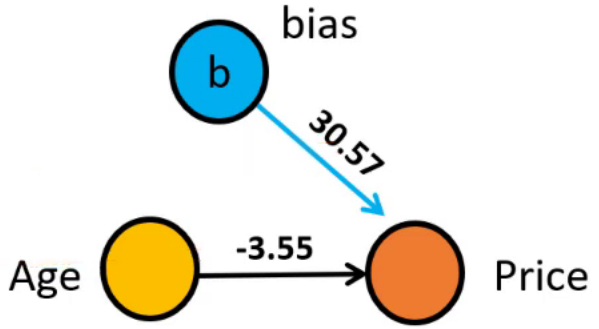
ANN vs Linear regression



$$z = w_0 + w_1 \cdot x_1$$

$$f(z) = z$$

$$f(z) = 30.57 - 3.55 \cdot x_1$$



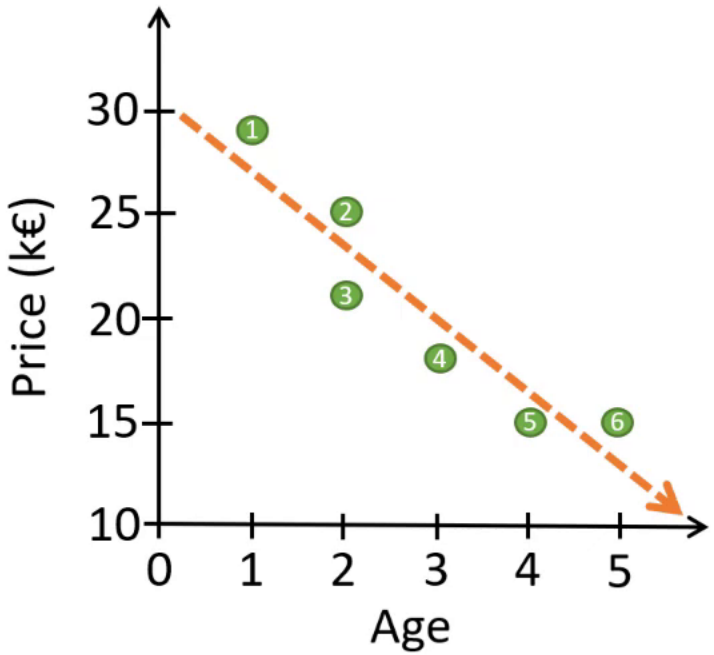
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



Age = 3

like this.

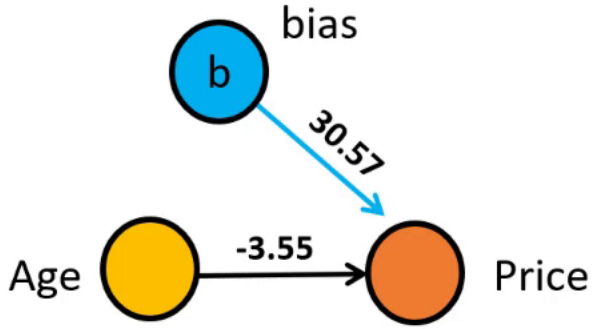
ANN vs Linear regression



$$z = w_0 + w_1 \cdot x_1$$

$$f(z) = z$$

$$f(z) = 30.57 - 3.55 \cdot x_1$$



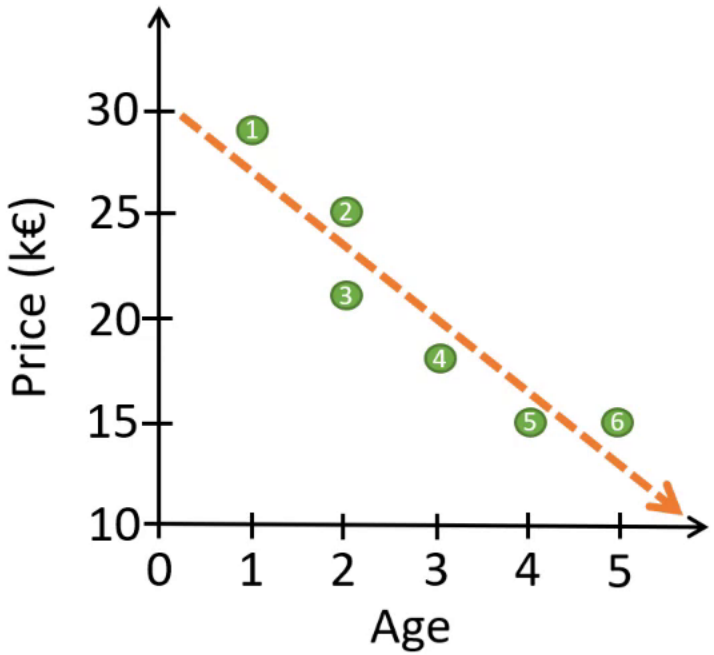
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



Age = 3

Note that this is exactly the same equation we used in linear regression to estimate the price.

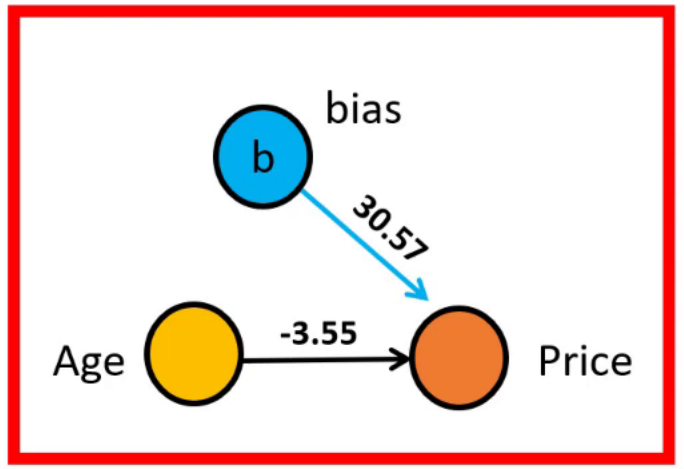
ANN vs Linear regression



$$z = w_0 + w_1 \cdot x_1$$

$$f(z) = z$$

$$f(z) = 30.57 - 3.55 \cdot x_1$$



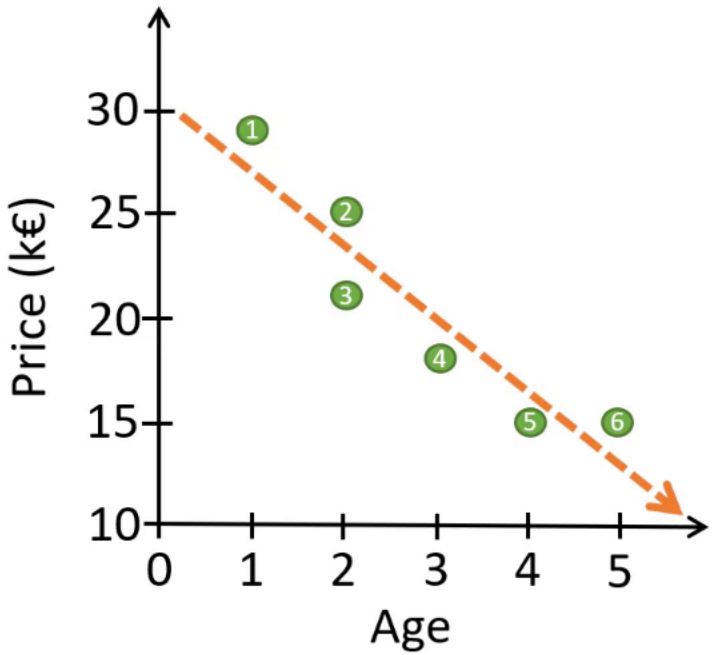
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



Age = 3

We can therefore conclude that this simple neural network, without a hidden layer, and with the identity function,

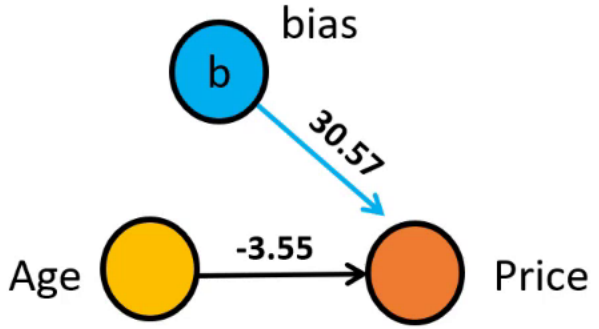
ANN vs Linear regression



$$z = w_0 + w_1 \cdot x_1$$

$$f(z) = z$$

$$f(z) = 30.57 - 3.55 \cdot x_1$$



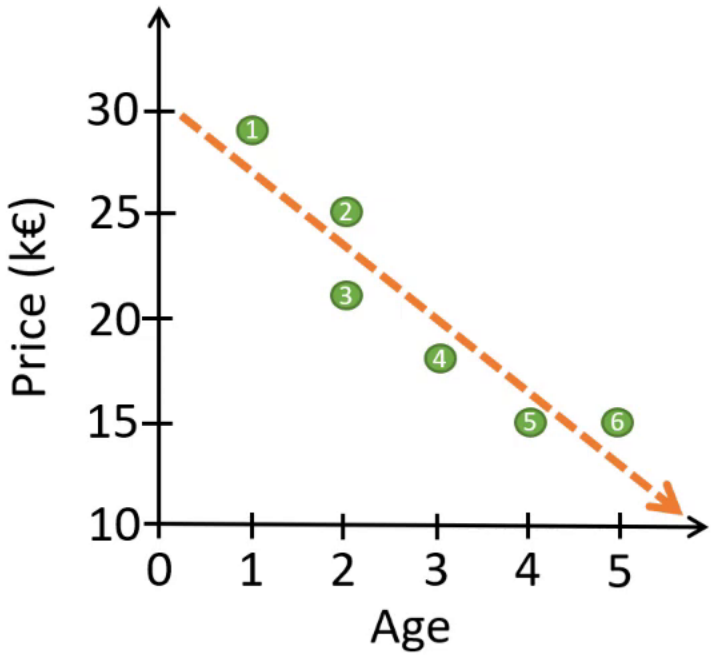
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$



Age = 3

is exactly the same thing as linear regression in this case.

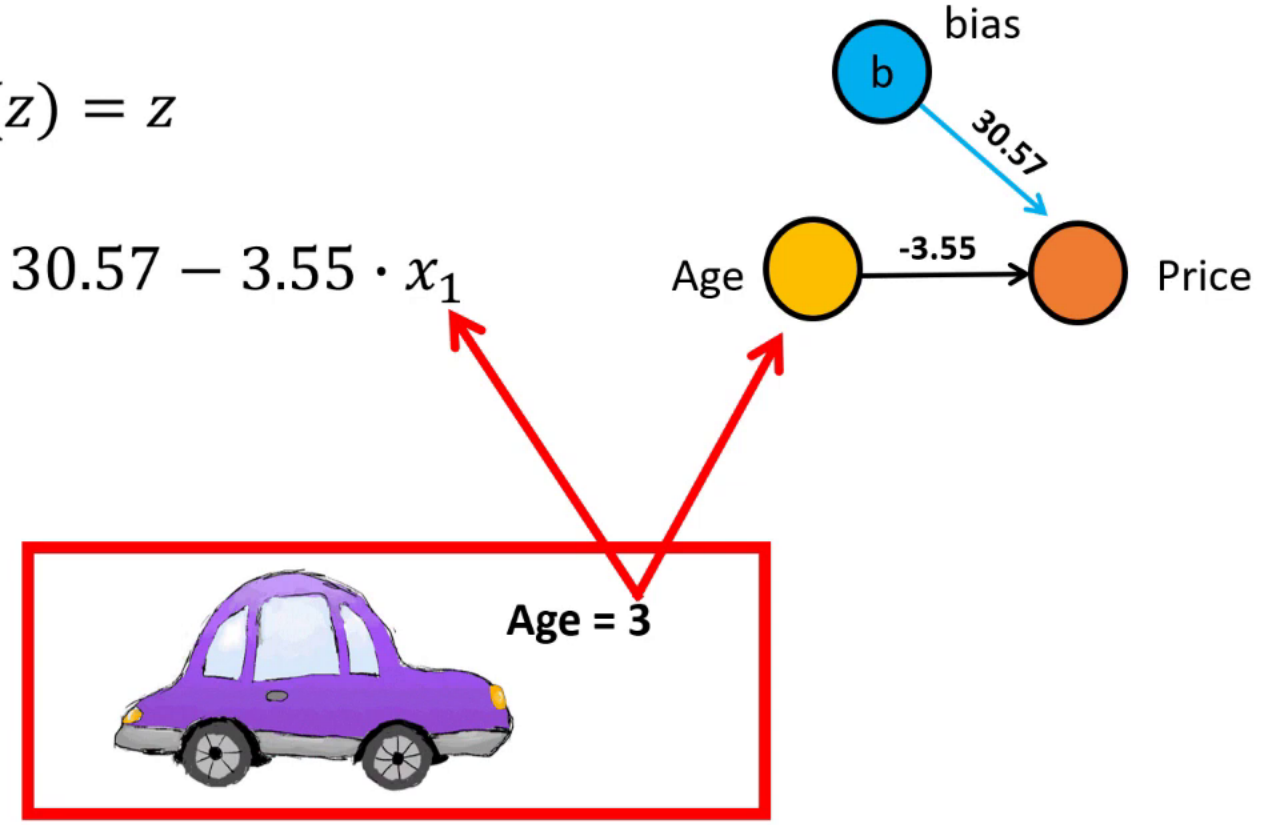
ANN vs Linear regression



$$z = w_0 + w_1 \cdot x_1$$

$$f(z) = z$$

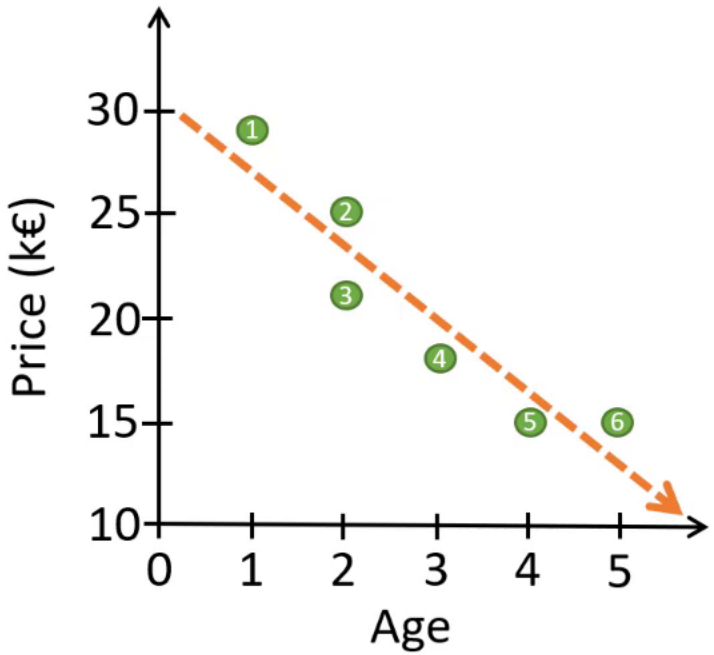
$$f(z) = 30.57 - 3.55 \cdot x_1$$



$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$

To predict the price of the car using the network, we plug in the value in the input node,

ANN vs Linear regression

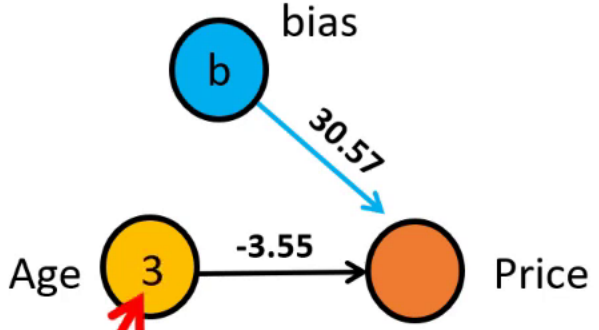


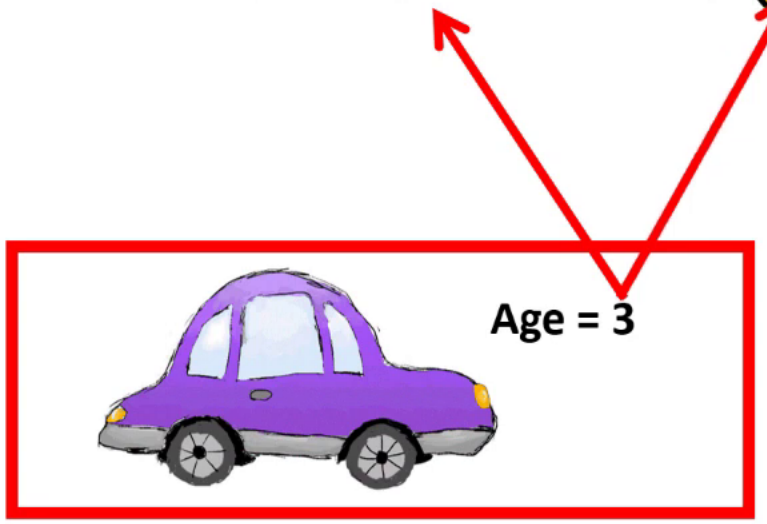
$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$

$$z = w_0 + w_1 \cdot x_1$$

$$f(z) = z$$

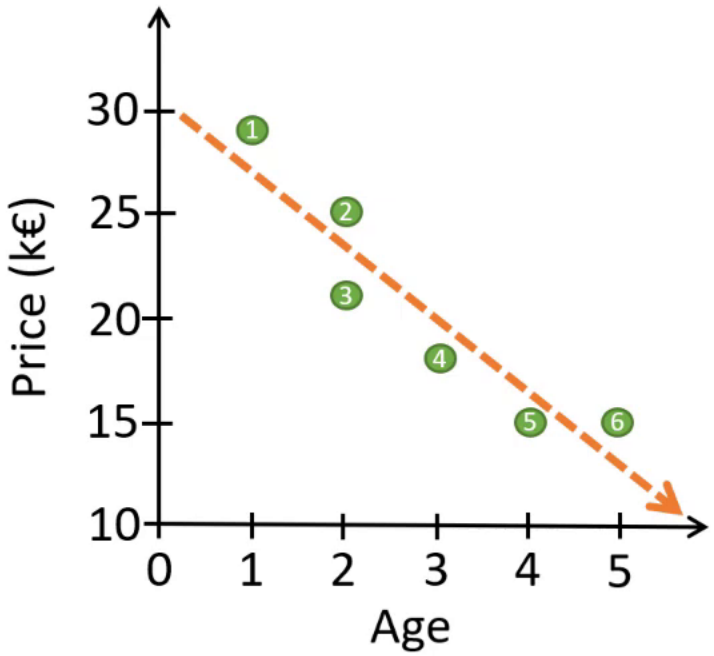
$$f(z) = 30.57 - 3.55 \cdot 3$$





like this,

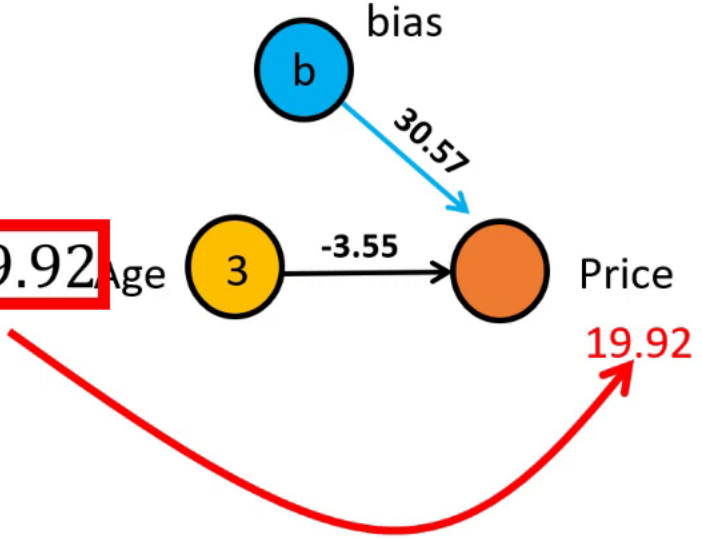
ANN vs Linear regression



$$z = w_0 + w_1 \cdot x_1$$

$$f(z) = z$$

$$f(z) = 30.57 - 3.55 \cdot 3 = 19.92$$



$$\text{Price} = 30.57 - 3.55 \cdot \text{Age}$$

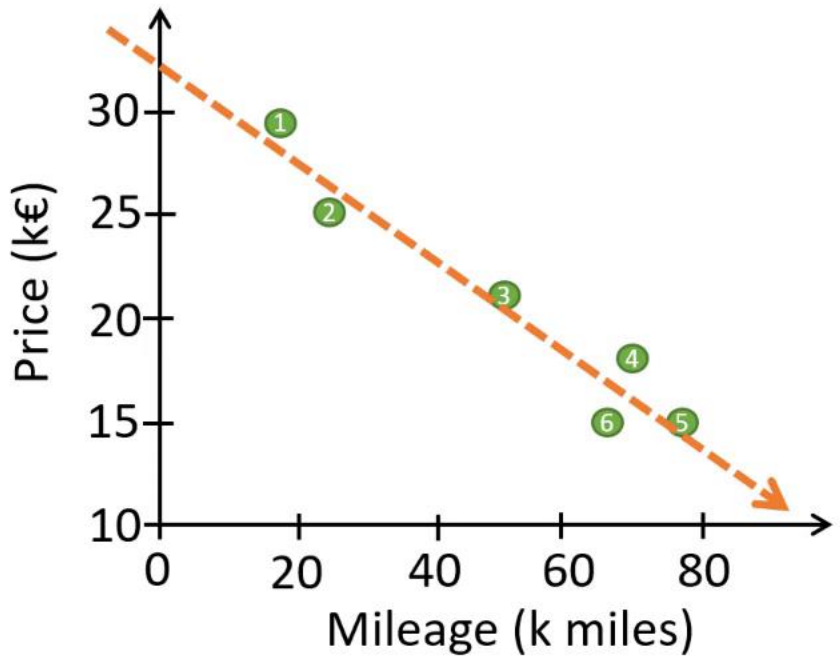


Age = 3

and do the math. We see that the network predicts the exact same price as the linear regression model.

ANN vs Linear regression

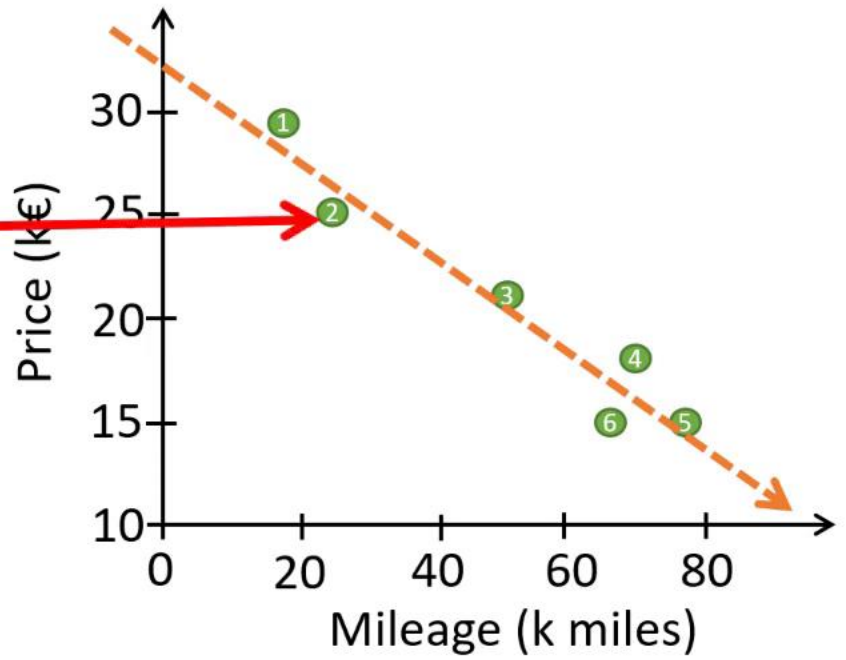
Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65



Now, suppose that we also have information about the mileage.

ANN vs Linear regression

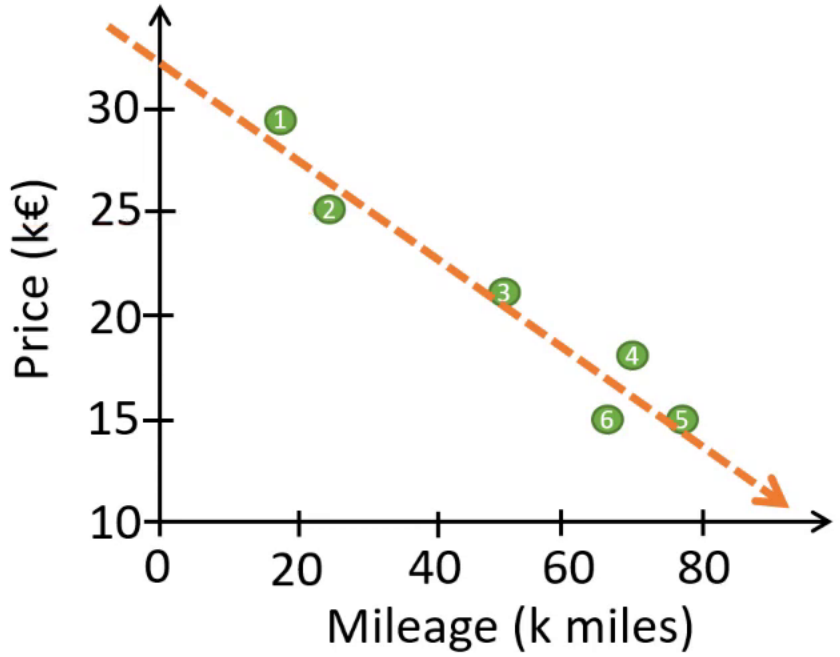
Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65



For example, car number two has been driven 25 thousand miles.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65

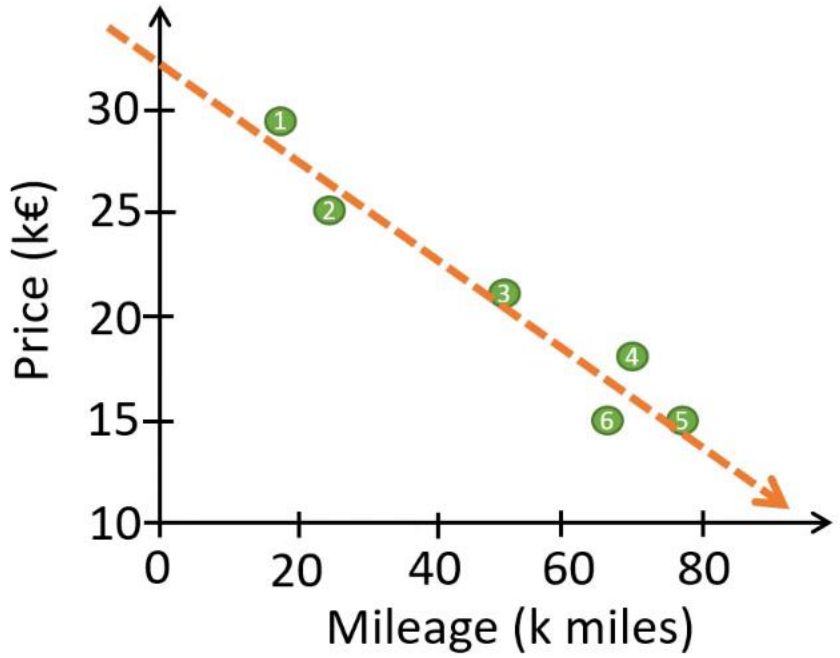


$$\text{Price} = 32.46 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage}$$

By using multiple linear regression, we can fit the following model to the data,

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65

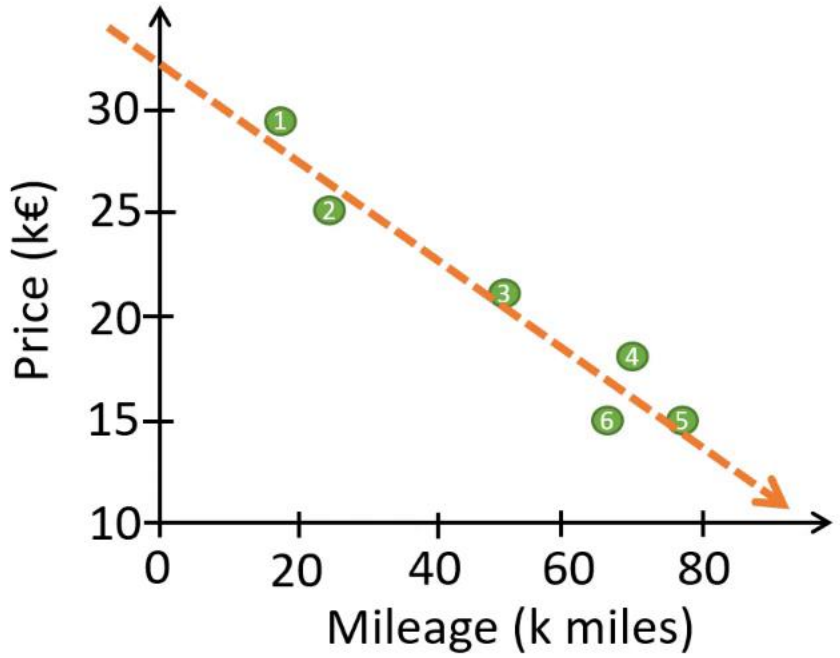


$$\text{Price} = 32.46 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage}$$

where the intercept has now been estimated to 32.46.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65

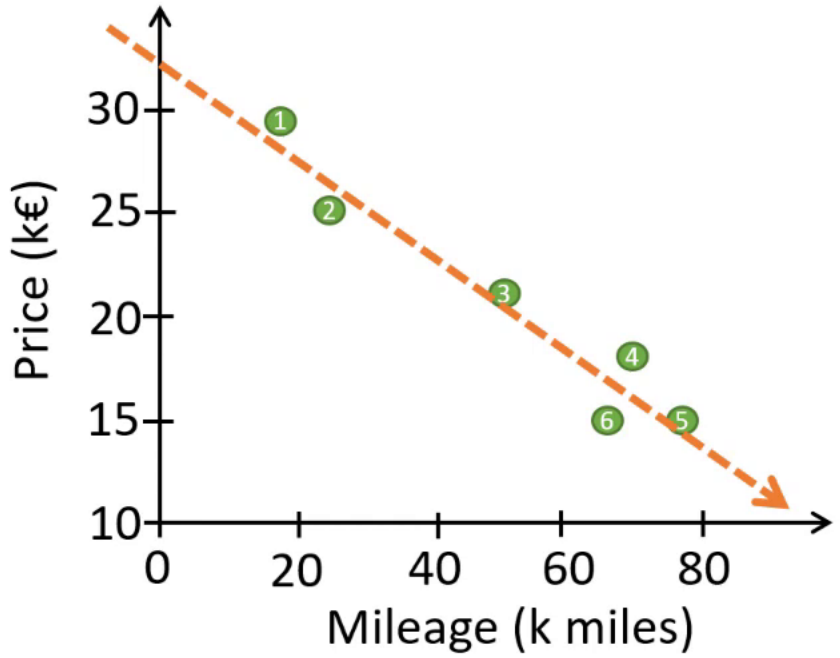


$$\text{Price} = 32.46 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage}$$

When the mileage is constant, then the price is reduced by 1540 euros each year. This could be seen as how much the price is expected to fall each year if a car is not used.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65

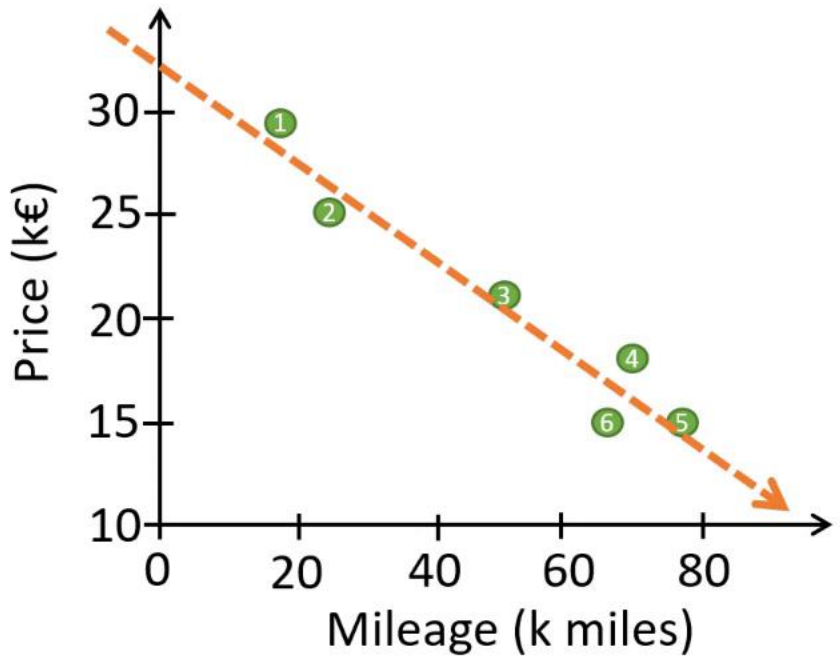


$$\text{Price} = 32.46 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage}$$

Even though we do not use the car, the price will still decline by 1.540 euros every year simply because the car gets older.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65

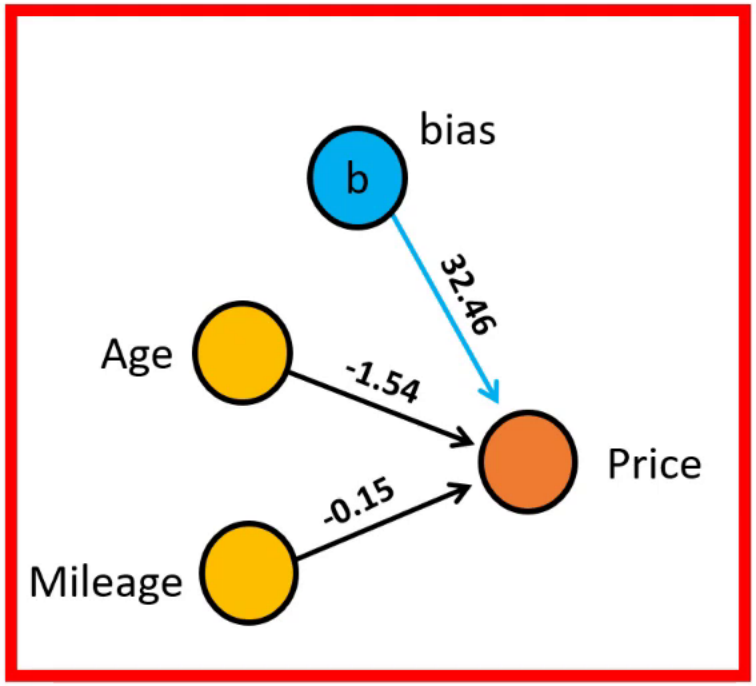


$$\text{Price} = 32.46 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage}$$

For example, if we compare cars of the same age, the difference in price between these cars is then dependent on the mileage. If the age is kept constant, the price is reduced by 150 euros for every thousand miles.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65

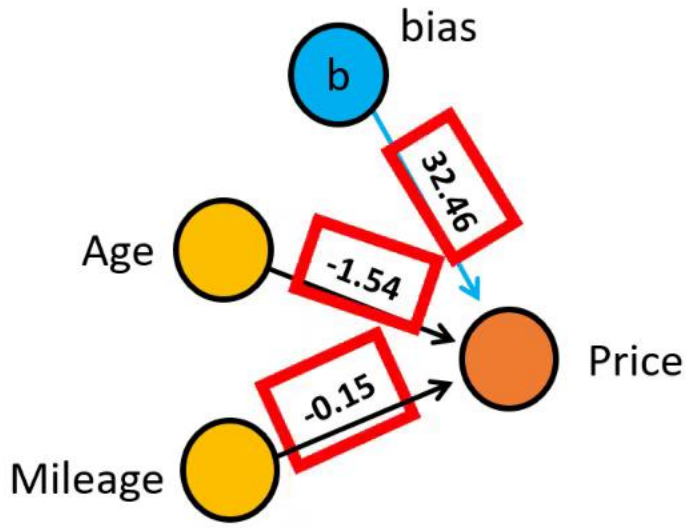


$$\text{Price} = 32.46 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage}$$

The corresponding neural network would look something like this,

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65

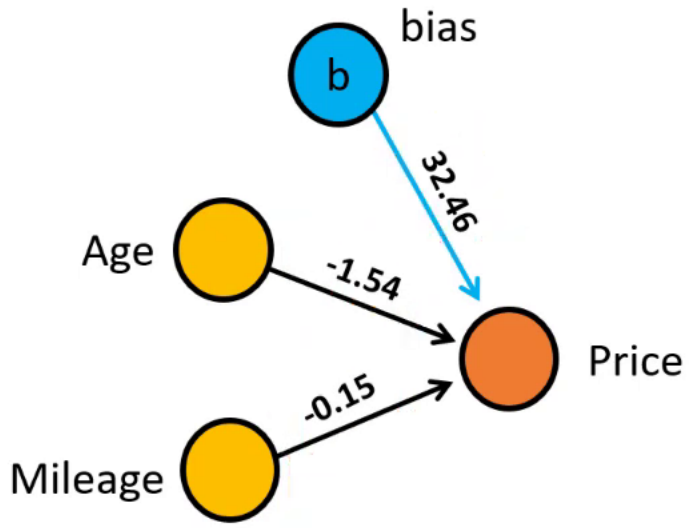


$$\text{Price} = 32.46 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage}$$

The values of the weights,

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65

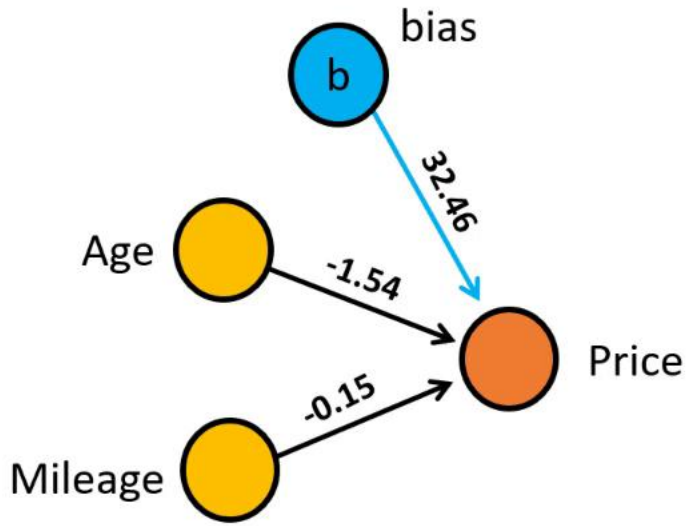


$$\text{Price} = 32.46 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage}$$

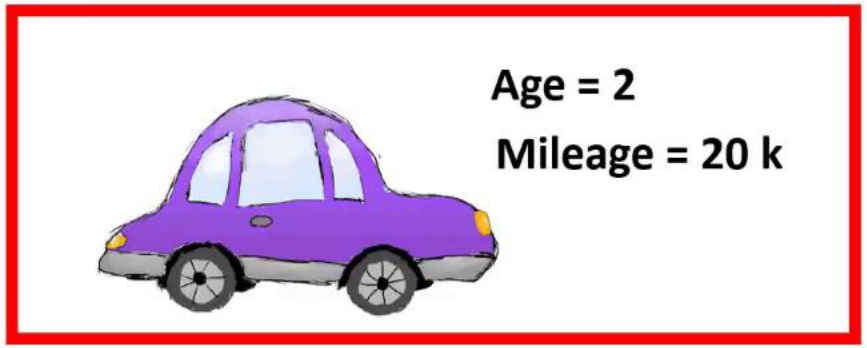
correspond exactly to the estimated parameters of the linear regression model.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65



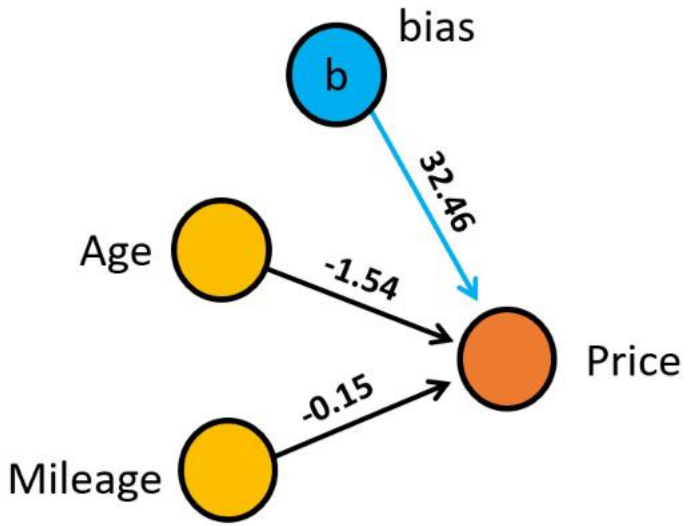
$$\text{Price} = 32.46 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage}$$



Suppose that we like to estimate the price of the following car,

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65



$$\text{Price} = 32.46 - 1.54 \cdot 2 - 0.15 \cdot 20$$



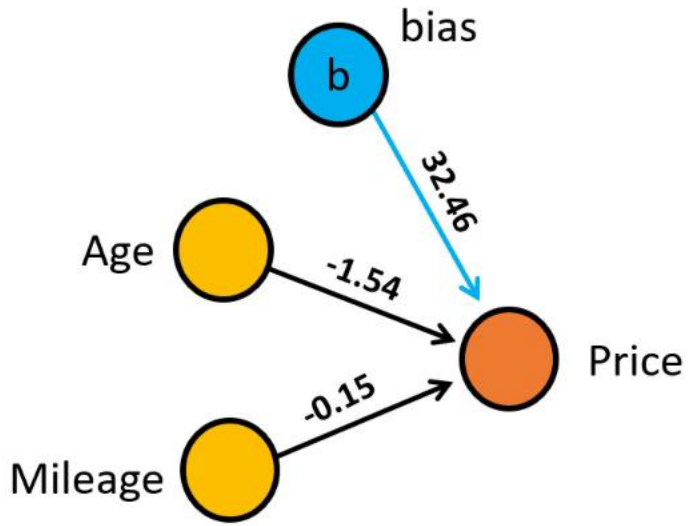
Age = 2

Mileage = 20 k

we therefore plug in the age and the mileage in the equation,

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65



$$\text{Price} = 32.46 - 1.54 \cdot 2 - 0.15 \cdot 20 = 26.38$$



Age = 2

Mileage = 20 k

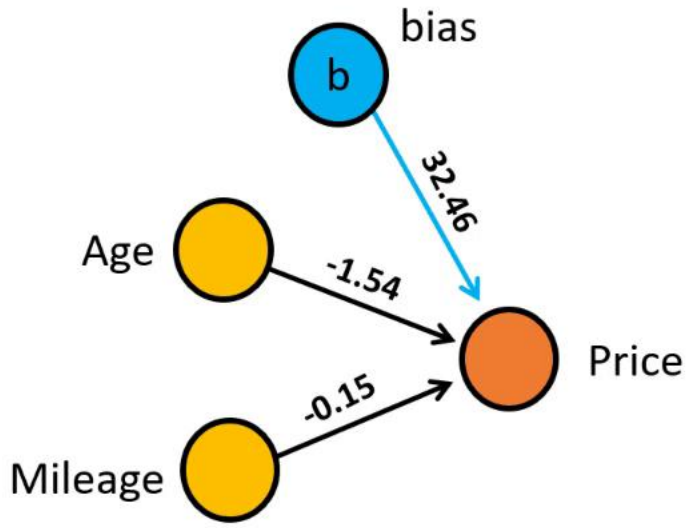
and do the math. We see that the model predicts the price to about 26.380 euros.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65

$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = z$$



$$\text{Price} = 32.46 - 1.54 \cdot 2 - 0.15 \cdot 20 = 26.38$$



Age = 2

Mileage = 20 k

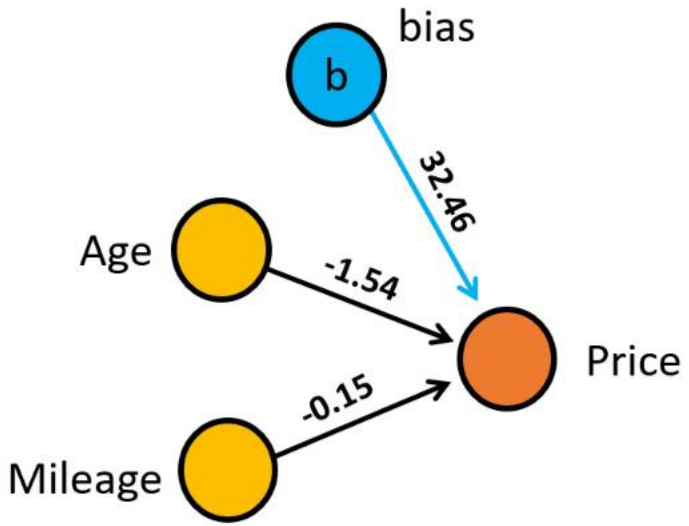
We can calculate the same thing with a neural network.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65

$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = z$$



$$\text{Price} = 32.46 - 1.54 \cdot 2 - 0.15 \cdot 20 = 26.38$$



Age = 2

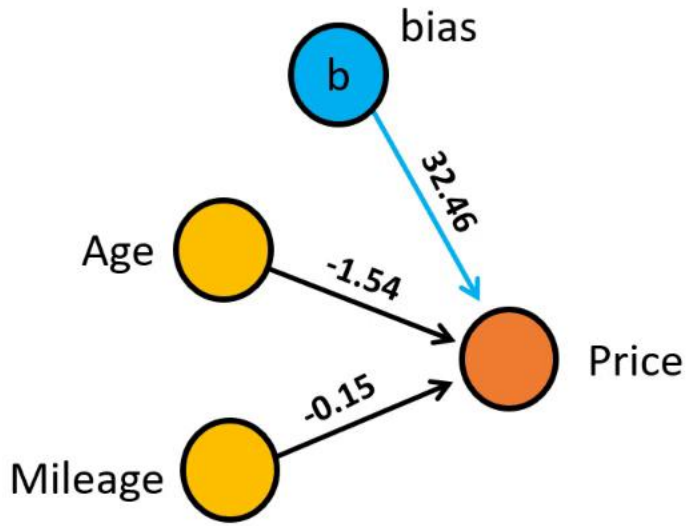
Mileage = 20 k

Since we in this case have two input nodes that are associated with the output node, n is here equal to two,

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65

$$z = w_0 + w_1 \cdot x_1 + w_2 \cdot x_2$$



$$f(z) = z$$

$$\text{Price} = 32.46 - 1.54 \cdot 2 - 0.15 \cdot 20 = 26.38$$



Age = 2

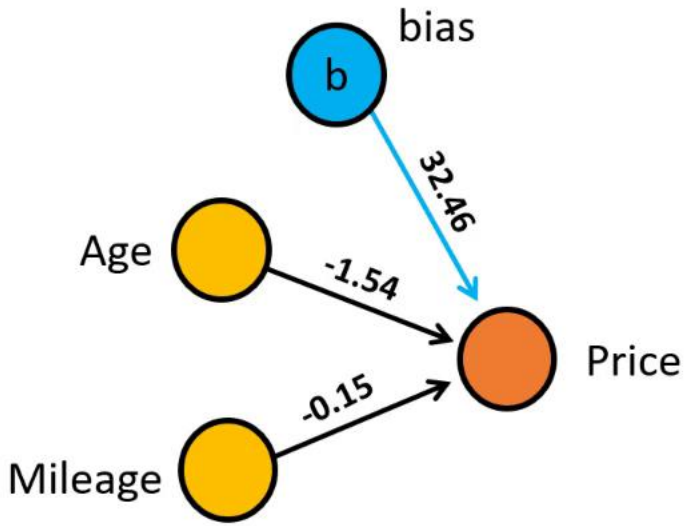
Mileage = 20 k

which means that we have the following equation.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65

$$z = 32.46 - 1.54 \cdot x_1 - 0.15 \cdot x_2$$



$$f(z) = z$$

$$\text{Price} = 32.46 - 1.54 \cdot 2 - 0.15 \cdot 20 = 26.38$$



Age = 2

Mileage = 20 k

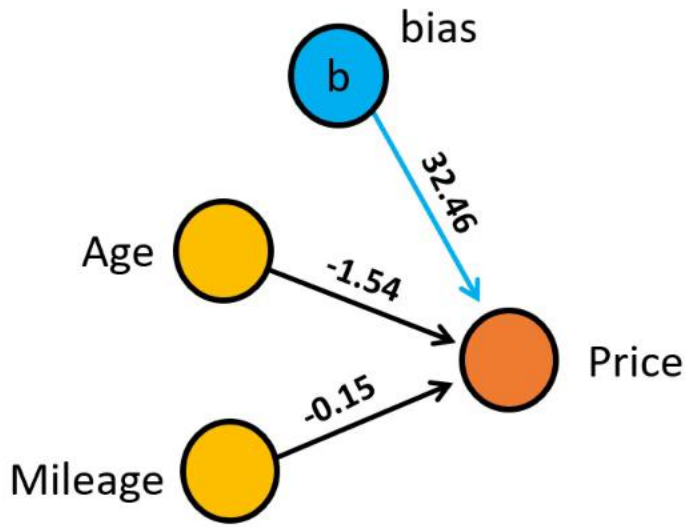
If we plug in the weights,

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65

$$z = 32.46 - 1.54 \cdot 2 - 0.15 \cdot 20 = 26.38$$

$$f(z) = z$$



$$\text{Price} = 32.46 - 1.54 \cdot 2 - 0.15 \cdot 20 = 26.38$$



Age = 2

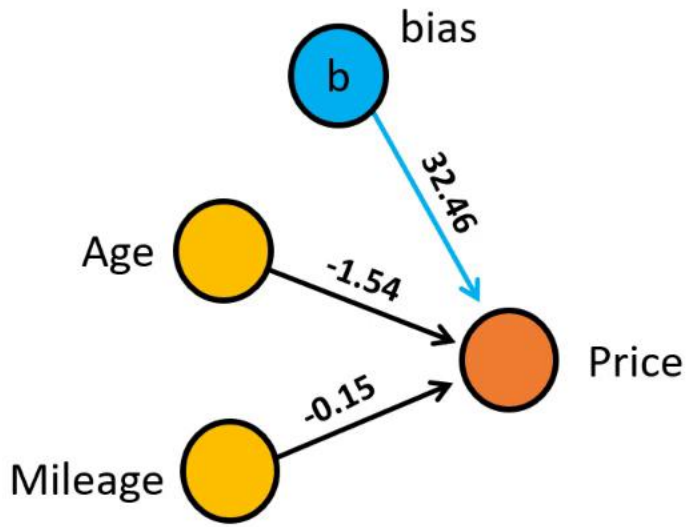
Mileage = 20 k

and do the math, and plug in this value in the identity activation function,

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65

$$z = 32.46 - 1.54 \cdot 2 - 0.15 \cdot 20 = 26.38$$



$$f(z) = 26.38$$

$$\text{Price} = 32.46 - 1.54 \cdot 2 - 0.15 \cdot 20 = 26.38$$



Age = 2

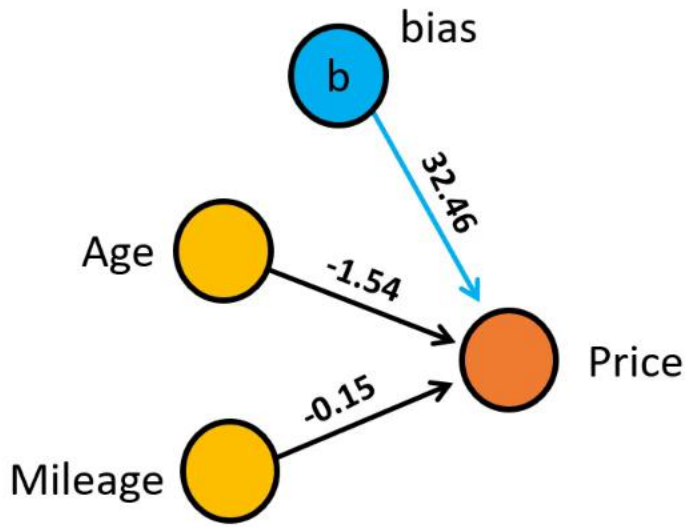
Mileage = 20 k

we see that the value of the output node is equal to 26.38,

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)
1	29	1	18
2	25	2	25
3	21	2	50
4	18	3	68
5	15	4	75
6	15	5	65

$$z = 32.46 - 1.54 \cdot 2 - 0.15 \cdot 20 = 26.38$$



$$f(z) = 26.38$$

$$\text{Price} = 32.46 - 1.54 \cdot 2 - 0.15 \cdot 20 = 26.38$$



Age = 2

Mileage = 20 k

which is exactly the same value as the predicted value of our regression model.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)	Color
1	29	1	18	Purple
2	25	2	25	White
3	21	2	50	Purple
4	18	3	68	White
5	15	4	75	White
6	15	5	65	Purple



Note that we can also include a categorical variable in our model, such as type of color.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)	Color
1	29	1	18	Purple
2	25	2	25	White
3	21	2	50	Purple
4	18	3	68	White
5	15	4	75	White
6	15	5	65	Purple



In this example, car number one, three and six have purple color,

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)	Color
1	29	1	18	Purple
2	25	2	25	White
3	21	2	50	Purple
4	18	3	68	White
5	15	4	75	White
6	15	5	65	Purple



whereas car number two, four and five have white color.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)	Color
1	29	1	18	1
2	25	2	25	0
3	21	2	50	1
4	18	3	68	0
5	15	4	75	0
6	15	5	65	1



Let's code the purple color as ones,

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)	Color
1	29	1	18	1
2	25	2	25	0
3	21	2	50	1
4	18	3	68	0
5	15	4	75	0
6	15	5	65	1



and the white color as zeros.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)	Color
1	29	1	18	1
2	25	2	25	0
3	21	2	50	1
4	18	3	68	0
5	15	4	75	0
6	15	5	65	1

$$\text{Price} = 32.41 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage} + 0.06 \cdot \text{Color}$$



If we fit a multiple linear regression model to the data, we will obtain the following estimated parameters.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)	Color
1	29	1	18	1
2	25	2	25	0
3	21	2	50	1
4	18	3	68	0
5	15	4	75	0
6	15	5	65	1

$$\text{Price} = 32.41 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage} + 0.06 \cdot \text{Color}$$



The variable Color in this equation takes the value one if the car has a purple color, or zero if the car is white.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)	Color
1	29	1	18	1
2	25	2	25	0
3	21	2	50	1
4	18	3	68	0
5	15	4	75	0
6	15	5	65	1

$$\text{Price} = 32.41 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage} + 0.06 \cdot \text{Color}$$



This value can therefore be interpreted as a car with purple color has a predicted price that is generally 0.06 thousand euros, or 60 euros greater than a white car with the same age and mileage.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)	Color
1	29	1	18	1
2	25	2	25	0
3	21	2	50	1
4	18	3	68	0
5	15	4	75	0
6	15	5	65	1

$$\text{Price} = 32.41 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage} + 0.06 \cdot \text{Color}$$



Age = 2
Mileage = 20 k
Color = 1

Suppose that we would like to use the model to predict the price of the following car,

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)	Color
1	29	1	18	1
2	25	2	25	0
3	21	2	50	1
4	18	3	68	0
5	15	4	75	0
6	15	5	65	1

$$\text{Price} = 32.41 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage} + 0.06 \cdot \text{Color}$$



Age = 2

Mileage = 20 k

Color = 1

with purple color.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)	Color
1	29	1	18	1
2	25	2	25	0
3	21	2	50	1
4	18	3	68	0
5	15	4	75	0
6	15	5	65	1

$$\text{Price} = 32.41 - 1.54 \cdot \boxed{2} - 0.15 \cdot \boxed{20} + 0.06 \cdot \boxed{1}$$



Age = 2

Mileage = 20 k

Color = 1

Then we plug in the corresponding numbers here,

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)	Color
1	29	1	18	1
2	25	2	25	0
3	21	2	50	1
4	18	3	68	0
5	15	4	75	0
6	15	5	65	1

$$\text{Price} = 32.41 - 1.54 \cdot 2 - 0.15 \cdot 20 + 0.06 \cdot 1 = 26.44$$



Age = 2

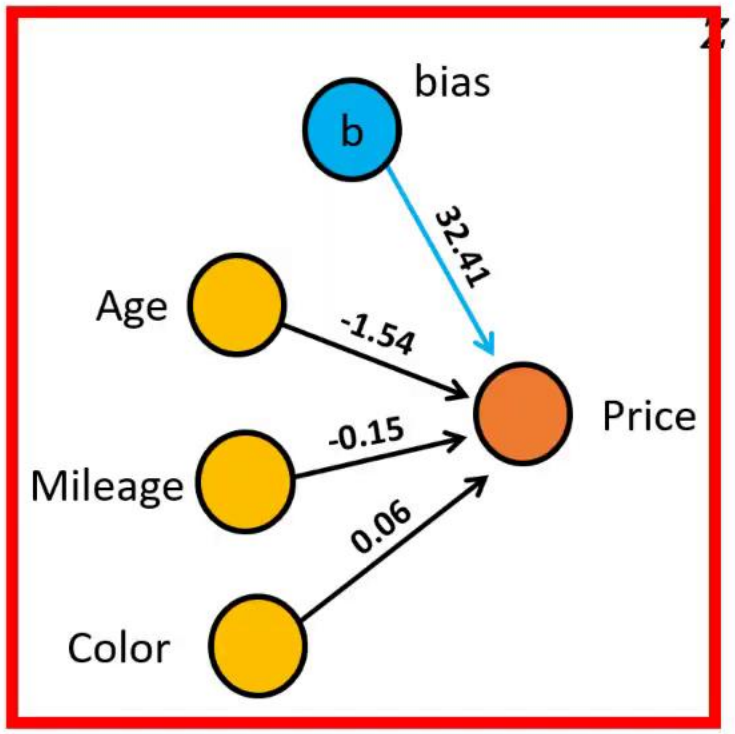
Mileage = 20 k

Color = 1

and do the math. The model predicts the price to 26.440 euros. Note that this calculation involves the rounded values of the estimated parameters.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)	Color
1	29	1	18	1
2	25	2	25	0
3	21	2	50	1
4	18	3	68	0
5	15	4	75	0
6	15	5	65	1



$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = z$$

$$\text{Price} = 32.41 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage} + 0.06 \cdot \text{Color}$$



Age = 2

Mileage = 20 k

Color = 1

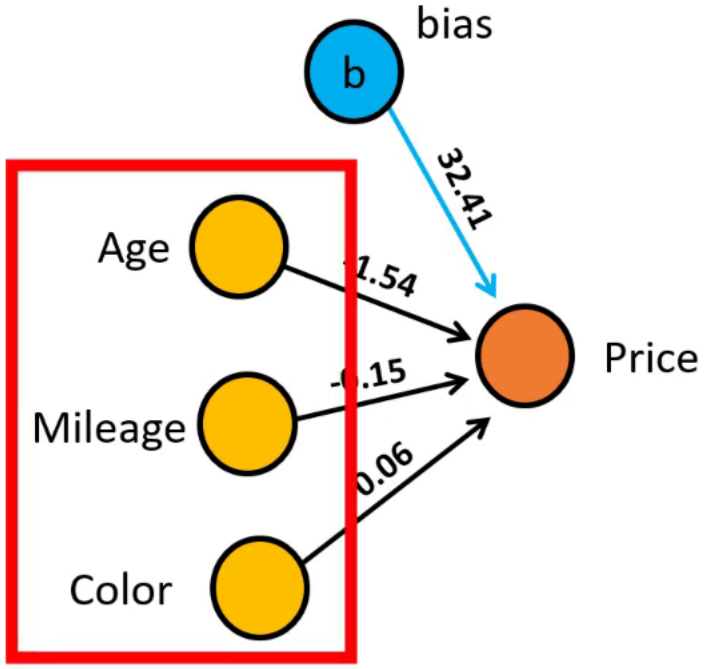
The corresponding neural network would look like this,

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)	Color
1	29	1	18	1
2	25	2	25	0
3	21	2	50	1
4	18	3	68	0
5	15	4	75	0
6	15	5	65	1

$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = z$$



$$\text{Price} = 32.41 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage} + 0.06 \cdot \text{Color}$$



Age = 2

Mileage = 20 k

Color = 1

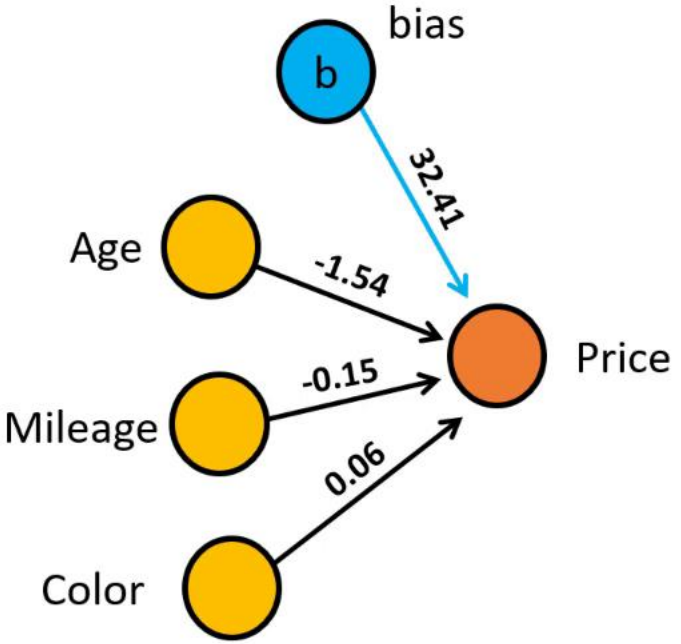
where we now have three input nodes.

ANN vs Linear regression

Car	Price (k€)	Age (years)	Mileage (k miles)	Color
1	29	1	18	1
2	25	2	25	0
3	21	2	50	1
4	18	3	68	0
5	15	4	75	0
6	15	5	65	1

$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = z$$



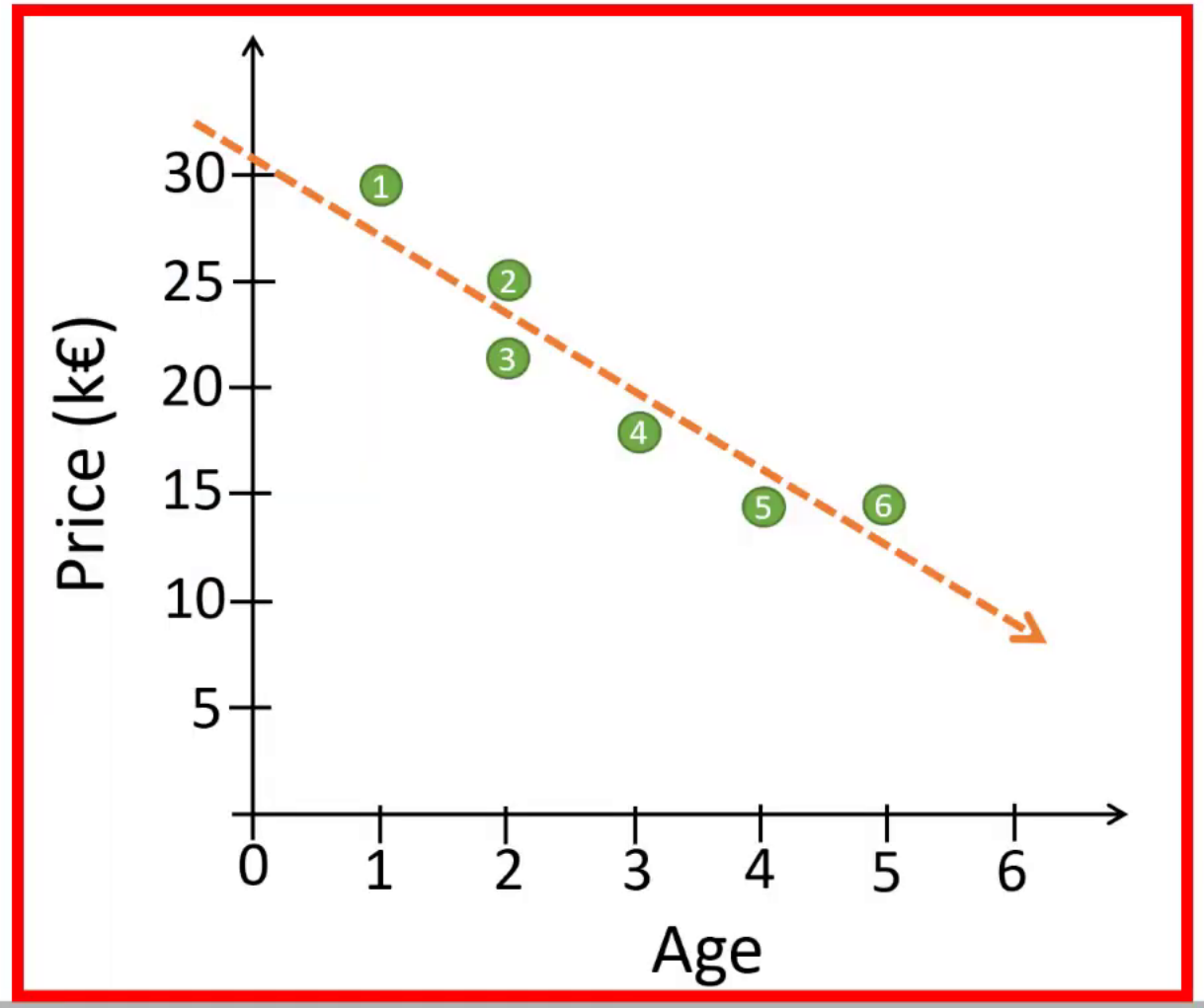
$$\text{Price} = 32.41 - 1.54 \cdot \text{Age} - 0.15 \cdot \text{Mileage} + 0.06 \cdot \text{Color}$$



Age = 2
Mileage = 20 k
Color = 1

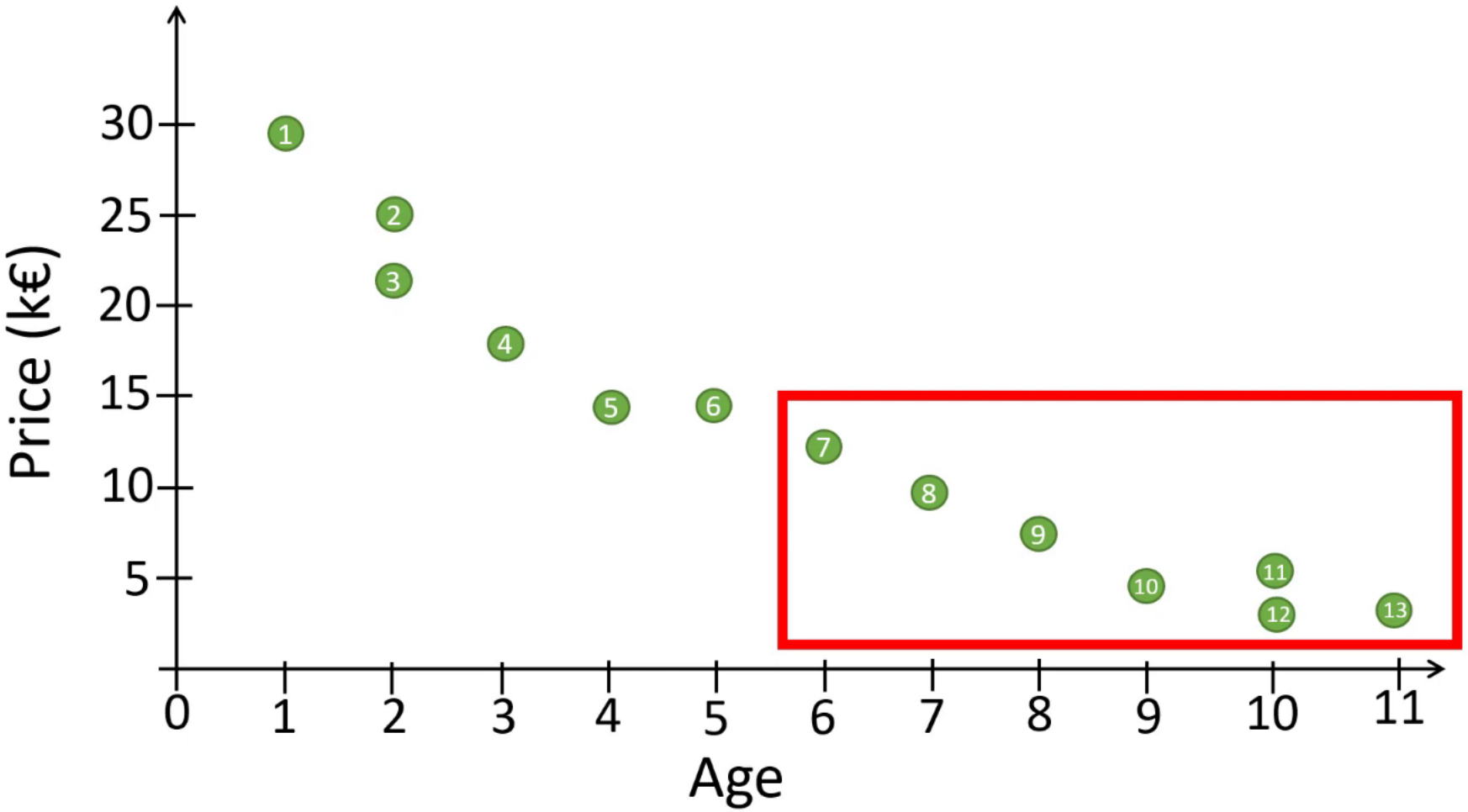
Using this network to predict the price of the following car will result in the exact same predicted price as in the regression model.

ANN vs Linear regression



One problem with linear regression is that we have to assume that the price falls according to a straight line, which means that the predicted price falls with the same amount every year.

ANN vs Linear regression



If we would collect the price on older cars, we would probably see that the price starts to approach zero like this.